



TUD | The Collaborative University

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Excellence Strategy of the Federal and State Governments (2019–2026)
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First self-assessment report for the evaluation of TUD | Dresden University of Technology, 2025

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First Self-Assessment Report
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TUD|Dresden University of Technology 2025

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Dresden, August 1, 2025

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Glossary and Abbreviations

Abbreviation	Explanation
AI	Artificial Intelligence
BiC	Behaviour in Context: Brain and Machine Computations of Behaviour in Complex and Uncertain Environments (TUD Excellence Cluster proposal [2026-32] that was not funded)
BMFTR	Federal Ministry for Research, Technology and Space (Bundesministerium für Forschung, Technologie und Raumfahrt)
BMWE	Federal Ministry for Economic Affairs and Energy (Bundesministerium für Wirtschaft und Energie)
BOSCH	German multinational engineering and technology company
boOst	Business Opportunities East (Ost) GmbH (one of 10 BMFTR-funded startup factories; TUD is a major partner)
CARE	Climate-Neutral and Resource-Efficient Construction (TUD Excellence Cluster 2026-32)
CASUS	Centre for Advanced Systems Understanding
CDIO	Chief Digitalisation and Information Management Officer
CeTI	Centre for Tactile Internet with Human-in-the-Loop (TUD Excellence Cluster 2019-25; 2026-2032)
cfaed	Centre for Advancing Electronics Dresden (TUD Excellence Cluster 2012-19)
CFLab	Construction Future Lab (TUD Campus Lusatia Research Infrastructure funded by Fund for Structural Change in German Coal Regions)
CIDS	Centre for Interdisciplinary Digital Sciences (TUD Science and Resource Centre)
CircEcon	Research Centre for Climate-Neutral Circular Economy (TUD Campus Lusatia Research Infrastructure; Joint Academic Unit of TUD, TU Chemnitz, TU Bergakademie Freiberg, and University of Applied Science Zittau-Görlitz; funded by Fund for Structural Change in German Coal Regions)
CMCB	Centre for Molecular and Cellular Bioengineering (TUD Central Academic Unit)
CODIP	Centre Open Digital Innovation and Participation (CIDS department)
Come2Data	Competence Centre for Interdisciplinary Data Sciences (3rd party-funded; CIDS department)
CoMos	Computational Modelling and Simulation (CIDS department)
CRC/TRR	DFG-funded Transregional Collaborative Research Centre
CRTD	Centre for Regenerative Therapies Dresden (TUD Excellence Cluster 2006-19)
ctd.qmat	Complexity, Topology and Dynamics in Quantum Matter (TUD Excellence Cluster 2019-25; 2026-32)

Abbreviation	Explanation
DAAD	German Academic Exchange Service (Deutscher Akademischer Austauschdienst)
DALI	Dresden Advanced Light Infrastructure
DDc	The legal entity DRESDEN-concept Science and Innovation Campus e.V. is an alliance of 43 leading research (Max Planck Institutes, Leibniz Institutes, Helmholtz Centres and Institutes, Fraunhofer Institutes), academic and cultural institutions in and around Dresden
DFG	German Research Foundation (Deutsche Forschungsgemeinschaft)
DIDS	Data-Intensive and Digital Science
DIGS-ILS	Dresden International Graduate School for Interdisciplinary Life Sciences
DIW Econ	Consulting company of the German Institute for Economic Research (Deutsches Institut für Wirtschaftsforschung, DIW)
DZA	German Centre for Astrophysics: Research, Technology, Digitalisation (Deutsches Zentrum für Astrophysik)
DZNE	German Center for Neurodegenerative Diseases (Deutsches Zentrum für Neurodegenerative Erkrankungen) (DDc partner)
EA	Excellence Area
EB	Executive Board
ECR	Early Career Researcher
EEG	Electroencephalogram
EF	Emerging Field
EKFZ	Else Kröner Fresenius Centre for Digital Health (Else Kröner Fresenius Zentrum für Digitale Gesundheit; Academic Unit TUD Medical Faculty)
EMBL	European Molecular Biology Laboratory
ERC	European Research Council
ESMC	European Semiconductor Manufacturing Company
est.	established
ETH Zurich	Federal Institute of Technology Zurich (Eidgenössische Technische Hochschule Zurich)
EU	European Union
EXC	Cluster of Excellence
EXCITELab Accelerator	Dresden-based high-tech accelerator founded in cooperation with the Dresden University of Technology that focuses on connecting high-tech startups with established corporations
EXU	University of Excellence
FIS	Major national research infrastructure (ForschungsInfraStruktur)
GlobalFoundries	Multinational company and leading manufacturer of semiconductors

Abbreviation	Explanation
GPT	Generative pre-trained transformer: an artificial neural network used in natural language processing
hd-tDCS	High-Definition Transcranial Direct Current Stimulation
HERITAGE network	First network between India and Europe for education and research cooperation in engineering sciences
HPC	High-Performance Computing
HR	Human Resources
HZDR	Helmholtz Centre Dresden-Rossendorf (Helmholtz-Zentrum Dresden-Rossendorf) (DDc partner)
iGEM	International Genetically Engineered Machine: a worldwide synthetic biology competition
IHI Zittau	International University Institute Zittau (Internationales Hochschulinstitut Zittau) (TUD Central Academic Unit)
IIT	Indian Institute of Technology
IITM	Indian Institute of Technology Madras
IGNITE	Internationally operating scale-up accelerator for early-stage deep tech startups founded by Intel
Infineon Technologies	Largest microcontroller manufacturer in the world, largest German-owned semiconductor manufacturer
IP	Intellectual property
IRTG	DFG-funded International Research Training Group
ISC	TUD International Strategy Council
KCL	King's College London (transCampus partner)
KPI	Key Performance Indicator
LAB	Living Art of Building (German Research Institute for the future of construction, in formation) (TUD Campus Lusatia)
MIT	Massachusetts Institute of Technology
MPI CBG	Max Planck Institute for Cellular Biology and Genetics (DDc Partner)
MRI	Magnetic Resonance Imaging
New Work	New Work focuses on transforming traditional work practices, including work space arrangements, to enhance flexibility, autonomy and collaboration
NFDI	National Research Data Infrastructure (Nationale Forschungsdateninfrastruktur)
NHR	National High Performance Computing (Nationales Hochleistungsrechnen)
OLED	Organic Light-Emitting Diode
PoL	Physics of Life (TUD Excellence Cluster 2019-25; 2026-32)
Q1 Publications	Top 25% of journals in a particular field based on Impact Factor

Abbreviation	Explanation
RefLau	Reference Power Plant Lusatia (BMW-fund project to establish an innovative power plant that uses renewable energies to demonstrate the possibilities of sector coupling)
REC ²	Responsible Electronics in the Climate-Change Era (TUD Excellence Cluster 2026-32)
RTG	DFG-funded Research Training Group
RWTH Aachen	RWTH Aachen University (Rheinisch-Westfälische Technische Hochschule Aachen)
ScaDS.AI	Centre for Scalable Data Analytics and Artificial Intelligence (BMFTR and DAAD funded)
SCD	Support Centre Digitalisation (CIDS Department)
SDG	Sustainable Development Goals
SECAI	School of Embedded Composite Artificial Intelligence (BMFTR- and DAAD-funded; one of three in Germany)
SEMECO	Secure Medical Microsystems and Communications (BMFTR Cluster4Future)
SI	Serbski Institute (TUD Affiliated Institute)
SLICE-DE	Digital Research Infrastructure for Data Processing and Communication Services (Digitale Forschungsinfrastruktur für Datenverarbeitung und Kommunikation)
SML	Smart Mobility Lab (TUD Campus Lusatia Research Infrastructure funded by Fund for Structural Change in German Coal Regions)
SLO	Saxon Science Liaison Office
SLUB	Saxon State and University Library (Sächsische Landesbibliothek – Staats-und Universitätsbibliothek)
STEM	Science, Technology, Engineering, and Mathematics
Stifterverband	Association of around 3,500 companies, foundations, science organisations and individuals that work to rethink and reshape science and education in Germany
STIPT	Semiconductor Talent Incubation Programme Taiwan
SynoSys	Synergy of Systems Centre (CIDS Department)
tFUS	Transcranial Focused Ultrasound
TGFS	Equity investor for knowledge-based, technology-oriented startups in Saxony (TechnologieGründerFonds Sachsen Management GmbH)
THE	Times Higher Education
TMS	Transcranial Magnetic Stimulation
TSMC	Taiwan Semiconductor Manufacturing Company
TUD	Technische Universität Dresden; Dresden University of Technology
TUD excite	TUD Centre for Excellence in Innovation, Transfer and Entrepreneurship

Abbreviation	Explanation
TUD GA	TUD Graduate Academy
TUD InnoX Campus	TUD Innovation Campus with the X reflecting transfer and excellence
TUDAG	TU Dresden Aktiengesellschaft (est. 2000) TUD's private business; a joint stock company that supports knowledge and technology transfer from science to the private sector
TUDiSC	TUD Disruption and Societal Change Centre
TZD	Technology Centre Dresden Ltd. German Law (Technologie-Zentrum Dresden GmbH) (TUD is a shareholder)
UCSD	University of California San Diego
UFZ	Helmholtz Centre for Environmental Research (Helmholtz-Zentrum für Umweltforschung)
UKD	University Hospital Dresden (Universitätsklinikum Dresden)
Unicorn	Privately-owned startup company valued at over US \$1 billion
University School	Joint project of the City of Dresden and TUD. The University School enables research on the digitalisation of primary education and project-based learning, and aims to educate student teachers and teachers.
von ardenne	Dresden-based German mechanical engineering company offering vacuum coating systems and technologies for functional layers with applications in research and industry
ZEISS	German manufacturer of optical systems and optoelectronics

German Abstract

Die TUD hat sich seit der Wiedervereinigung zu einer internationalen Spitzenuniversität entwickelt. Seit 2012 hat es uns der Exzellenzstatus ermöglicht, strategisch zu investieren, herausragende Talente zu gewinnen und Transformationskraft zu entfalten. Seit 2019 erlaubt uns die Dauerförderung als Exzellenzuniversität unser Profil weiterzuentwickeln und uns an internationalen Spitzenstandards zu messen. Die gemeinsame Reflektion unserer Entwicklung hat unser Selbstverständnis geschärft: TUD - The Collaborative University: inventive. transformative. engaged.

Die Integration von grundlagenbasierter und transformationsorientierter Forschung ist Motor unseres Erfolgs. Fünf Exzellenzcluster (2026–33) mit fächerübergreifender Spitzenforschung zeugen von unserem Leistungsvermögen. Mit den Exzellenzbereichen Principles of Life: Transforming Health Care, Matter, Advanced Materials: Transforming Structures und Microelectronics, Information Sciences: Transforming AI haben wir ein unverwechselbares Forschungsprofil geschaffen. Die in der TUD tief verwurzelte Kultur kollaborativer Interdisziplinarität werden wir durch Potentialbereiche wie „Responsible Mobility as a Resource“ und „Forest-Water Dynamics in Climate Change“, einen globalen Hub für Komplexitätsforschung und die institutionelle Integration von Medizin und Lebenswissenschaften weiter entfalten.

Die Forschungsinfrastruktur haben wir zusammen mit unseren Partnern der DRESDEN-concept Allianz weiter gestärkt. Neue Technologieplattformen, Labore und Supercomputer machen uns attraktiv für internationale Forschungspersönlichkeiten. Mit dem TUD|Campus Lausitz schaffen wir einen neuartigen Forschungs- und Transferstandort und gestalten Strukturwandel. Der Ausbau des Centers für interdisziplinäre digitale Wissenschaften und die Umsetzung neuer PPP-Modelle in Sachsen werden als Innovationstreiber wirken.

Im Technologietransfer nimmt die TUD eine Führungsrolle in Deutschland ein. Mittels der Exzellenzförderung haben wir mit TUD|excite ein Innovationszentrum geschaffen, das Transferaktivitäten bündelt und neue Maßstäbe in der Startup-Förderung setzt (z.B. Startup Factory boOst). Zukünftig werden thematisch fokussierte TUD|InnoX Campi die Innovationskette weiter beschleunigen.

Gesellschaftliches Engagement ist ein Signum unseres institutionellen Profils. Wir wirken durch das COSMO Science Forum und Programme wie „TUD im Dialog“ in die Gesellschaft und begeistern Jugendliche für Wissenschaft, besonders in den MINT-Fächern. Wir engagieren uns für eine offene und demokratische Gesellschaft. Künftig entwickeln wir wirkungsvolle Dialogformate weiter und bringen uns weiterhin als verantwortungsbewusste Stimme in kritischen Debatten ein.

Studium und Lehre haben wir enger an unsere Spitzenforschung gekoppelt und konsequent an internationalen Standards orientiert. Die Vermittlung von Kernkompetenzen oder projektbasiertes Lernen sind zentral in unserer Lernstrategie. Wir werden die Schwellen für Studierende ohne akademischen Hintergrund weiter abbauen, durch die Gestaltung und Nutzung von KI in der Lehre neue Wege gehen und durch Micro-Credentials lebensbegleitendes Studieren ermöglichen.

Enge Kollaboration mit unseren transCampus Partnern King's College London und IIT Madras und unseren Science Liaison Offices erhöhen unsere globale Reichweite. Die Zahl unserer internationalen Studierenden und Forschenden wächst stetig. Wir werden internationale Studierende gezielt auswählen, das transCampus-Model weitertragen, und unsere Zusammenarbeit im Dreiländereck Tschechien, Polen, Sachsen stärken.

Bei Personalentwicklung, Community-Building und Partizipation haben wir von internationalen Benchmarks gelernt: Beschleunigte Berufungsverfahren, passgenaue Förderung des wissenschaftlichen Nachwuchses, sowie modernisierte Governance-Strukturen machen uns agiler. Effektive Programme zur Gleichstellung, Diversität und Inklusion, die stetige Erhöhung des Frauenanteils auf allen Ebenen sowie ein gemeinsamer Wertekanon sind Ausdruck des Erfolges unserer Universitätskultur. Wir werden durch die konsequente Digitalisierung von Prozessen und strategisches Talentmanagement unsere Weiterentwicklung als lernende Organisation vorantreiben.

Neue digitale Technologien haben die Art, wie wir leben, lernen und produzieren fundamental verändert. Gleichzeitig erfordern geopolitische Disruption und die Gefährdung unserer Lebensgrundlagen von uns wissenschaftsbasierte Innovation und institutionelle Anpassungsfähigkeit. Als resiliente Forschungsuniversität entwickeln wir transformative Lösungen für die Herausforderungen des 21. Jahrhunderts und verbinden globale Wirkung mit regionaler Verantwortung. Die Herausforderungen sind groß für unseren Planeten, für die Demokratie und für unsere Rolle als Universität. Gut gerüstet durch unsere erprobten Stärken und ambitionierten Ziele, die wir mit der EXU-Förderung verfolgen können, sind wir bereit, ein neues Kapitel unserer Exzellenz zu schreiben.

Part 1: Progress Achieved within the Framework of the Funding, Results and Effects of the Approved Institutional Strategy

A. Overall Assessment: TUD's Journey of Excellence

Strategic Evolution and Vision

TUD stands for the power of collaboration driving institutional tenacity and inventiveness, both shaping and being shaped by profound social, political, and technological transformations. Our Vision is to engage and pioneer transformative solutions for the core challenges of the 21st century combining global impact with regional leadership. Since 2019, when our excellence status became permanent, we have leveraged EXU funding to amplify our successful trajectory. The key achievements (as linked with our 2018 EXU proposal) and the lessons learnt are explained in Chapters 1 to 9.

Research Excellence and Profile Development

Our research momentum is demonstrated by increased third-party funding: five Clusters of Excellence selected in 2025, ranking us third among German universities; increased top 25% journal (Q1) publications, and an outstanding Field-Weighted Citation Impact. Our EXCs drive internationally influential research as well as innovations. Together with streamlined appointment procedures they facilitate talent recruitment. Strategic investments in Emerging Fields such as automated mobility, neurosciences, and data-intensive science have yielded significant returns. Based on systematic analysis of coordinated research programmes, strengthened by DRESDEN-concept partners, we created three transformative Excellence Areas: "Principles of Life: Transforming Health Care", "Matter. Advanced Materials: Transforming Structures", and "Microelectronics. Information Sciences: Transforming AI".

Infrastructure, Research Ecosystem, and Research Support

Our progress has been backed by comprehensive infrastructure development and innovative research-support services. In collaboration with DRESDEN-concept Alliance partners, we established advanced technology platforms and research facilities, including those under construction on the newly founded TUD|Campus Lusatia, demonstrating our commitment to regional transformation and our ability to seize opportunities. We have improved our support for early-career researchers through devising multiple qualification tracks and professionalised human resources development, while fostering an equitable, inclusive, and open university culture through strategic investments in equal opportunities and diversity.

Teaching, Transfer, and Impact

We have advanced the research-teaching nexus by integrating independent junior group leaders and supporting student research initiatives. Establishing our Excellence Centre for Innovation, Transfer and Entrepreneurship (TUD|excite) has fostered significant achievements, such as procuring one of the 10 prestigious German Startup Factories as well as significant company partnerships. TUD ranks first among German universities for number of patents held. In 2024, we commissioned an independent study by DIW Econ (cf. [Glossary](#)) that quantified our economic impact, showing that every Euro of public funding from Saxony generates twofold returns and a substantial number of jobs. We have become a strong voice in science communication as well as for defending democratic values and institutions in public debate.

Drivers of Innovation: International Collaboration and Digitalisation

We strategically expanded our global impact through strengthening the transCampus with King's College London and establishing a new transCampus partnership and Saxon Science Liaison Offices in Taiwan and India, accompanied by increasing numbers of international scientific staff and international student enrolments. The creation of the Centre for Interdisciplinary Digital Sciences and the newly created Chief Office 'Digitalisation and Information Management' have accelerated our digital transformation across all operational domains.

Governance and Strategic Learning

Using EXU funding, we significantly enhanced our governance, including Executive Board reconfiguration, systematic School evaluation, strengthened strategy development, and advanced academic analytics. Our experience has highlighted the importance of strategic agility, particularly under constraining conditions, emphasising the need for accelerated digitalisation, refined financial modelling, and fostering institutional resilience.

Where Does This Leave Us?

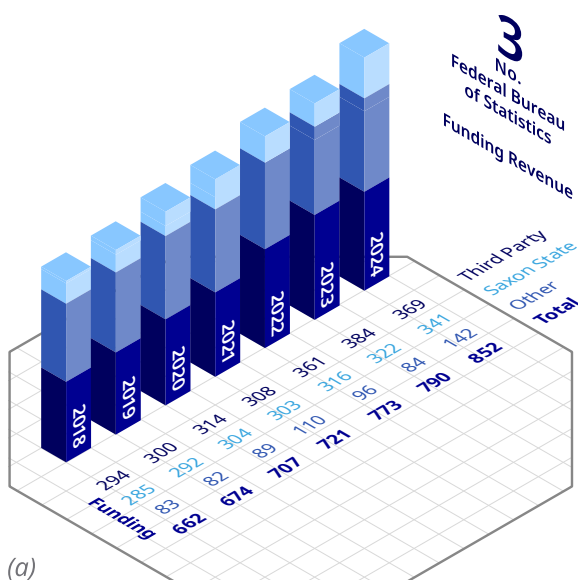
EXU funding has strengthened all of TUD – propelling research, enriching education, leading transfer to the next level, and motivating us to renew our strategy. Not least it contributed to Saxony becoming *the* hub for the semiconductor industry in Europe. We have learned that organisational success requires scalable solutions, a careful balance between innovation and continuity, and data-based monitoring for rapid adaptation. This successful development leaves us in a better position to push the frontiers of knowledge, provide transformative solutions for core challenges facing humankind, to lead in global higher education, and to drive positive change in our region.

B. Achievements and Lessons Learnt

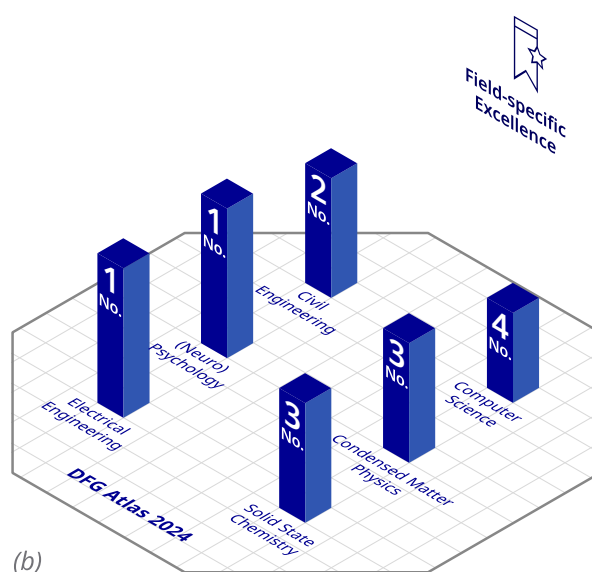
1. Research

In 2019, our overall goal was to institutionally facilitate **scientific excellence** and raise our international visibility. We are proud to report the following achievements: (i) elevated research visibility and **increased third-party funding** through key resources, (ii) **five EXC** awarded for 2026-32, (iii) substantive progress in five **Emerging Fields (EF)**.

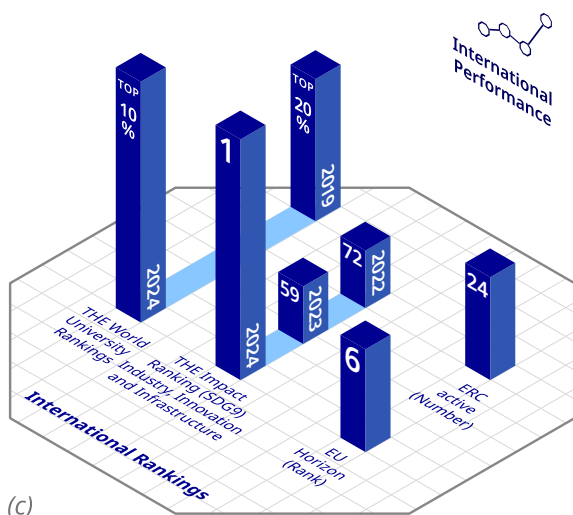
(i) Enhanced Visibility of Research Strength and Increased Third-Party Funding Through Key Resources



(a)



(b)



(c)

Figure 1. TUD's research excellence: (a) trajectory of third-party funding (cf. Annex: Tab. 1, 2.1), (b) field-specific excellence, (c) international rankings.

TUD's research excellence is reflected in its increasing third-party funding trajectory, with total **competitive funding** rising 23% from 2019 and 2024, securing **Germany's third-highest position** (Federal Bureau of Statistics; cf. Fig.1a). German Research Foundation (DFG) funding showed growth of 13.9% to €110M and TUD's **field-specific excellence** impresses through its commanding positions in the 2024 DFG Funding Atlas (cf. Fig.1b). Our international research capability is evidenced by our unmatched leadership in **DFG International Research Training Groups (IRTGs)**, complemented by our strongest-ever portfolio of 24 active European Research Council (ERC) grants - ranking sixth nationally - and our sixth position in number of TUD researchers funded by Horizon Eu-

rope among German universities (cf. Fig.1c). Not least, our collaborations with Max Planck, Helmholtz, Leibniz, and Fraunhofer partners within the DRESDEN-concept Alliance (DDc) made this success possible (cf. Fig.2).

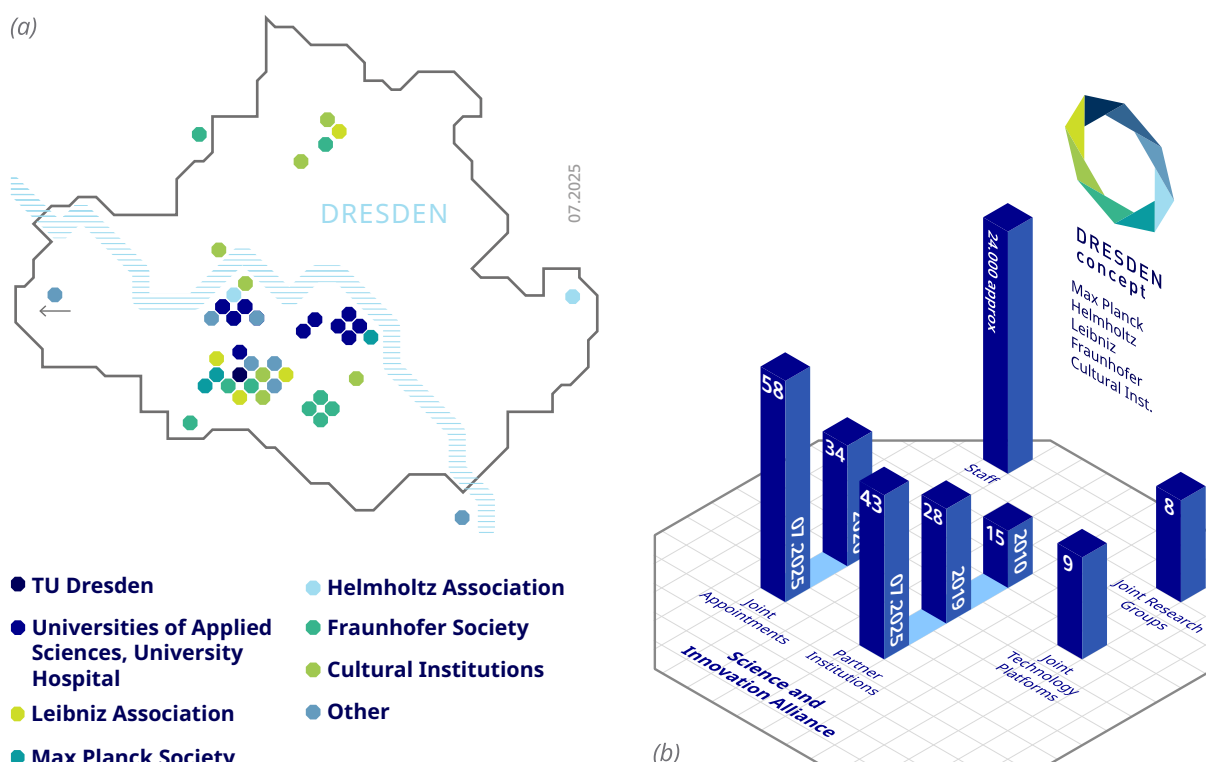


Figure 2. The DRESDEN-concept Science and Innovation Campus (DDc) is (a) a unique collaborative network of 43 leading research and cultural institutions in and around Dresden. (b) Based on joint appointments, joint technology platforms and DDc Research Groups, DDc plays a key role in TUD's success story.

Employing the criteria championed by the Declaration on Research Assessment and Coalition for Advancing Research Assessment, TUD's research quality metrics reveal remarkable momentum, with our **Field-Weighted Citation Impact** of 1.61 (2019-24) significantly exceeding benchmarks across both German and European Universities of Technology.

In 2019, TUD had **five research priority areas**: health sciences; information technology and microelectronics; materials science and engineering; energy and mobility; and culture and societal change. In line with our successful coordinated research programmes since 2019, we created a more distinctive research profile consisting of **three Excellence Areas (EAs)**: (a) **Principles of Life: Transforming Health Care**, (b) **Matter. Advanced Materials: Transforming Structures**, and (c) **Microelectronics. Information Sciences: Transforming AI** (cf. ANNEX: TAB.11). Figure 3 illustrates how the EAs are embedded in our overall research approach 2027-33 (cf. OUTLOOK).

Our renewed **research support services** have delivered significant benefits. TUD **Project Scouts** guide first-time grant applicants and organise networking events, making a significant difference for Early Career Researchers (ECRs). **New grant writing workshops** provide ECRs with focused training to secure funding. The **Network & Idea Studios** improved their support for innovation and teamwork, giving students, researchers, and industry partners a place to develop and test ideas together.

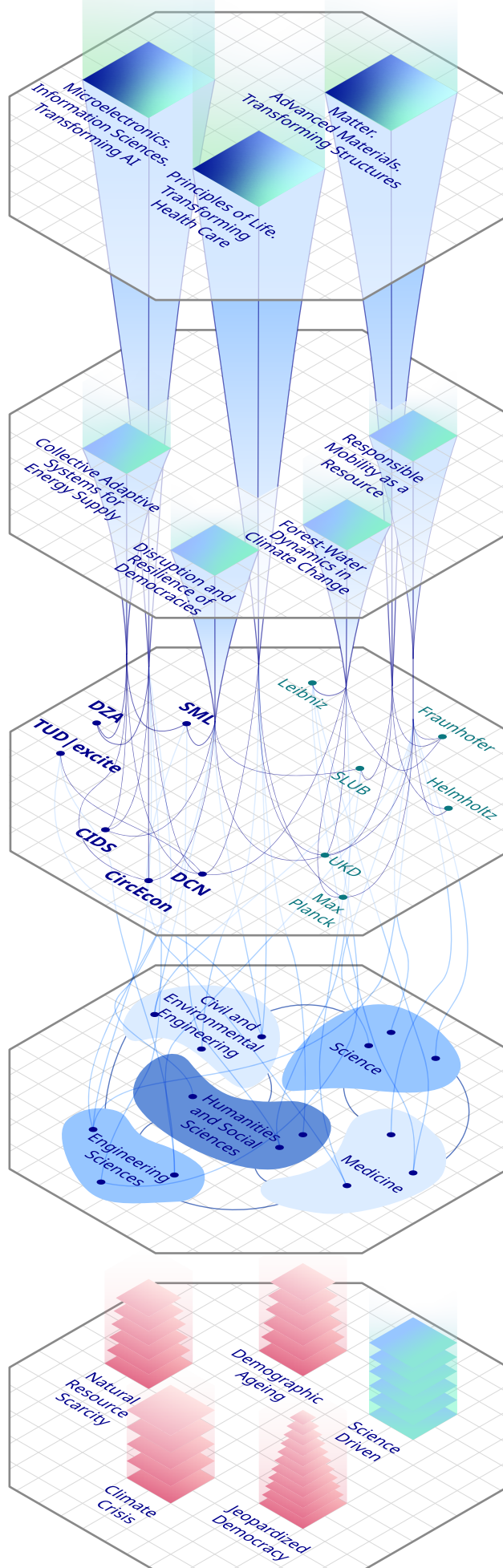
EXCELLENCE
AREAS

EMERGING
FIELDS

RESEARCH
ECOSYSTEM

SCHOOLS

MOTIVATION



DRESDEN-concept
Alliance

Figure 3. The TUD research approach and landscape (2027-33) is schematically illustrated by this five-tier model: Motivated by curiosity-driven research and key challenges of the 21st century, TUD researchers engage in interdisciplinary collaboration across Schools and are supported by a state-of-the-art research ecosystem (illustrated by select examples), as well as by the unique DDC Alliance. A mix of bottom-up and top-down activities leads to the creation of Emerging Fields which are an important source for our research excellence in the future.

Our **research-support system** provided over 2,350 consultations (2019-24) and our innovative workshop portfolio - spanning funding strategies, ideation, and pitch refinement - has been adopted by peer institutions. This comprehensive support framework has yielded tangible results: enhanced **ERC grant success**, increased **DFG individual grant funding** (2019-24: 8%), and maintaining a strong presence in DFG Collaborative Research Centres (CRCs; cf. [Annex: Tab.10](#)).

(ii) Five Successful EXC Applications

TUD excelled in procuring funding for **five EXCs**, placing us at the top of German Universities of Technology and number three in the overall ranking (cf. [Annex: Brief Profile](#)). All three existing EXCs - Physics of Life (PoL), Complexity, Topology and Dynamics in Quantum Matter (ctd.qmat, with University of Würzburg), and Tactile Internet with Human-in-the-Loop (CeTI) - have been renewed. In addition, two new EXCs have been funded: CARE (with RWTH Aachen) for climate-neutral and resource-efficient construction, and REC² for responsible electronics in the era of climate change. Reflecting our Vision, all five EXCs connect **fundamental research** with **solutions to critical global challenges** (cf. [OUTLOOK](#)).

(iii) Development of Emerging Fields (EF; cf. Annex: Tab.11)

In 2018, five EFs were identified to strengthen TUD's research profile: (1) Data-Intensive and Digital Science, (2) Neurosciences, (3) Autonomous and Networked Mobility, (4) Water Research, and (5) Societal Change. Next, we report on our achievements.

Data-Intensive and Digital Science (DIDS)

In 2018, we selected DIDS based on our strength in high-performance computing (HPC) and in data-intensive science. Institutional support expanded the HPC infrastructure, broadened it to include artificial intelligence (AI) approaches, attracted new researchers with complementary expertise, and advanced interdisciplinary collaboration among TUD scientists using DIDS methods. Through strategic development, DIDS has attained international visibility (e.g., Flatiron Institute NYC, Chan-Zuckerberg Biohub San Francisco, Open Geospatial Consortium, Washington DC; Centre for Applied Internet Data Analysis, UCSD) and scholarly impact commensurate with TUD's areas of excellence, marking its successful evolution **from EF to research powerhouse**.

DIDS has fostered TUD's excellence by adding uniqueness to all EXC proposals, strengthening industry partnerships (influencing **TSMC's** Dresden investment), advancing **Master's programmes** in Data Science and Computational Modelling, establishing strong research ties with the Helmholtz Institute for Advanced Systems Understanding (**CASUS**), and elevating TUD's prominence in data science nationally and internationally.

TUD successfully created a unique profile in AI Hardware and Embedded AI, reflected in publications from the **Christian Mayr** and **Thomas Mikolajick** labs and the recently awarded DFG CRC/Transregio 404. The EF was also strengthened by national research-data infrastructure (NFDI) grants (**BASE4NFDI**, **NFDI4Earth**). DIDS serves as a basis for high-profile interdisciplinary research and attracted considerable third-party funding: the BMFTR- and DAAD-funded Graduate School of Embedded Composite Artificial Intelligence (**SECAI**; one

of 3 AI Graduate Schools in Germany); and the Centre for Scalable Data Analytics and Artificial Intelligence (**ScaDS.AI**; one of 6 National AI Competence Centres in Germany). Open GPT-X, the European answer to generative pre-trained transformer (GPT) language models, developed by ten partner organisations from business, science, and media (ca. €15M); or the DARWIN project (TUD: €1.36M) that brings together Rolls-Royce Germany, German Aerospace Center Braunschweig, and others for a deeper understanding of airplane turbomachines using AI design methods (cf. [Annex: Tab.11](#)). DIDS collaborates with the Else Kröner Fresenius Centre (EKfZ) for Digital Health, **5G++Lab** Germany, the **Barkhausen Institute** (a TUD spinout led by **Gerhard Fettweis**, Vodafone Professor of Mobile Communication Systems), and the BMBF Cluster4Future ‘Secure Medical Microsystems and Communications (**SEMECO**)’ to provide an outstanding environment for innovative and sustainable collaboration at Europe’s leading academic and industrial hub for microelectronics, communications engineering, and explainable AI.

In total, **DIDS third-party funding** (2019-24) amounts to **€161M**, which is matched by its scientific productivity (185 peer-review publications, of which 75% in Q1 journals, >100 conference proceedings since 2019). Most recently, DIDS helped to attract €40M funding for **biomedical AI** from the **Boehringer Ingelheim Foundation**, the Max Planck Society, and Saxony.

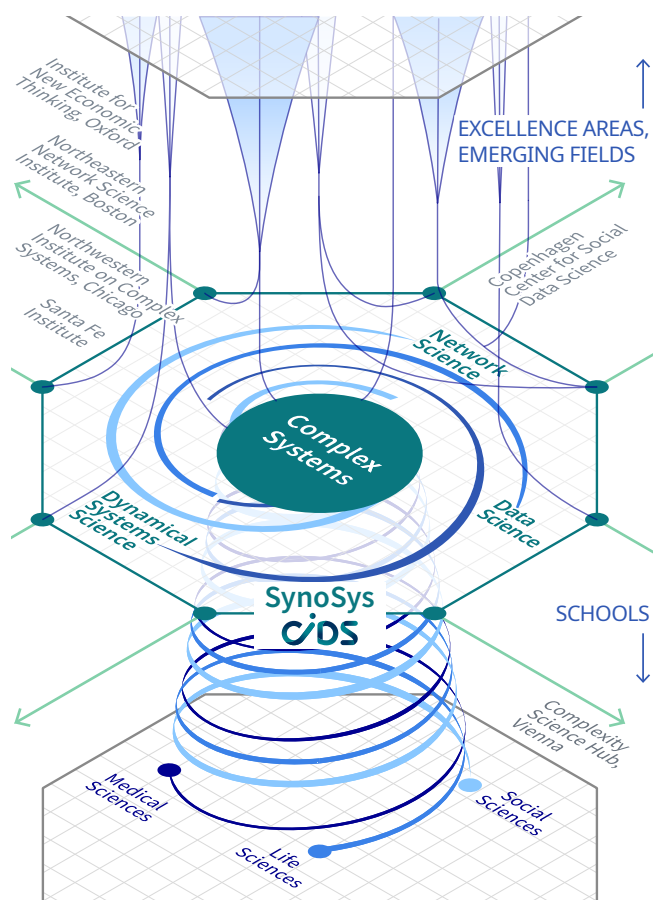


Figure 4. The Centre Synergy of Systems (SynoSys) is an important research department of CIDS and embedded in a world-wide network of Complexity Hubs (see also [Fig. 11](#)).

Based on DIDS’s success, TUD established a Science and Resource Centre for Interdisciplinary Digital Sciences (**CIDS**), which currently encompasses eight departments (cf. [DIGITALISATION: II; Fig.11](#)). **SynoSys**, one of the CIDS departments, applies transdisciplinary methods from a social-science, life-science and medical-science perspective (cf. [Fig.4](#)). It advances TUD’s mission by leveraging computational and network science to tackle pressing societal challenges, from pandemic preparedness to ecosystem stability and the effects of social media. In 2024, SynoSys successfully recruited **17 early career researchers (ECR)**, competing with leading US and European institutions. SynoSys’ first-year **funding success** (€3.5M secured, €3.2M pending) sets the stage for its advancement into a global complexity-science network hub (cf. [OUTLOOK: DRIVE1](#)).

Connecting with the humanities and social science, the CIDS department **Speculative Transformation** examines the societal impacts of digital transformation and is closely linked with the TUD Disruption and Societal Change Centre (TUDiSC) (cf. [RESEARCH: SOCIETAL CHANGE](#)).

Neurosciences

Selected as an EF for funding excellence, Neurosciences expanded through key appointments in Neuroscience (**Humboldt Professor Catherina Becker**, prior University of Edinburgh) and Biopsychology (**Bernhard Spitzer**, prior Max Planck Institute for Human Development), and through consolidation of **advanced research infrastructure** (including MRI, EEG, TMS, hd-tDCS, tFUS, eye tracking) (total invest since 2019: €3.2M).

EF Neurosciences' excellence has manifested in **prestigious grants** (transCampus TUD-King's College London (KCL) IRTG 2773, renewed CRC/TRR 940, CRC 205 and CRC/TRR 265), leadership in EXC initiatives (BiC, CeTI), and **outstanding metrics**: 54.5% Q1 publications (Ø 2019-24); Psychology Faculty's rise to 94th globally (THE Ranking), and top DFG funding nationwide (2024; cf. [Fig.1c](#)). TUD's Biosciences Research and Teaching were ranked 5th in Germany (Centre for Higher Education University Ranking 2024). The University Hospital together with the TUD Faculty of Medicine are among the top three Medical Centres in Germany.

This success is linked with a **growing collaborative network** - from core collaborations between TUD Faculties Medicine and Psychology, the TUD Centre for Regenerative Therapies Dresden (CRTD), and the German Centre for Neurodegenerative Diseases (DZNE) to interdisciplinary alliances with Computer Science and Transport Sciences within the EXC proposal Behaviour in Context (BiC). The **research ecosystem** encompasses prestigious institutions including the Max Planck Institute for Human Cognitive and Brain Sciences, the EKFZ for Digital Health, the German Centre for Child and Adolescent Health, Princeton's Neuroscience Department, and the Wyss Centre for Neuroscience, Geneva. Thus, strategic development has positioned the EF Neurosciences as a **catalyst for two of TUD's Excellence Areas** in healthcare transformation and in AI advancement (cf. [OUTLOOK: RES1](#)).

Autonomous and Networked Mobility

Integrating transport, social, and digital sciences, the EF Autonomous and Networked Mobility addresses emerging mobility challenges through interdisciplinary excellence. Strategic appointments, including **Excellence Professor Travis Waller** (prior University New South Wales) as well as **five international professorial appointments** since 2019, and **state-of-the-art facilities** (€130M investment: Vehicle Test Centre, TUD|Campus Südvorstadt, third-party funded Smart Mobility Lab (SML), TUD|Campus Lusatia; cf. [Fig.8](#)) position TUD internationally at the forefront of mobility transformation research. International partners include Delft University of Technology, University of Texas at Austin, and the Singapore-MIT Alliance for Research and Technology.

The EF's growth is demonstrated by **major DFG grants**, including the RTG 2947 for automated air transport (€10.9M) and CRC/TRR 408 'AgiMo' (€14.7M), major contributions to the **BiC EXC proposal** and scholarly impact, with Q1 journal representation more than doubling

since 2019 to 56.5% (2024), and is complemented by **Deutsche Bahn's endowed Chair** in Digital Railway Systems (cf. [Annex: Tab.12](#)).

The EF's trajectory demonstrates how global recruitments and interdisciplinary collaboration form the basis for success, while its evolution from technology-centric to human-centric approaches has proven essential in addressing the multifaceted challenges of automated mobility.

Water Research

The EF Water Research has harnessed the combined expertise of TUD and the Helmholtz Centre for Environmental Research (UFZ) to develop intelligent adaptation strategies that enhance water system resilience and mitigate the impact of water extremes on ecosystems and society.

We successfully appointed **Jakob Zscheischler** (prior ETH Zurich, Max Planck Institute Biogeochemistry) as Professor of Data Analytics in Hydro Sciences, complemented by two **Junior Research Groups** in Contaminant Science and Forest-Related Environmental Communication and a DDC research group on particle resuspension in environmental flows (jointly funded: Helmholtz-Zentrum Dresden-Rossendorf (HZDR, TUD)).

The Faculty of Environmental Sciences, housing the majority of the EF's principal investigators, has been renewed with **19 new professorial appointments** since 2019. It demonstrates strong **publication output**, with 75% of its publications (2019-24) appearing in Q1 journals. Additionally, the faculty's **third-party funding** has increased significantly, rising from €13.5M in 2019 to €18.4M in 2024. Further, the RTG 'Fostering Forest Systems F2S' has advanced to resubmission in 2025, alongside other coordinated proposals targeting water systems or forest resilience.

The EF is undergoing a strategic shift in its research focus: The recruitment of talented new faculty has sparked innovative research directions in **Forest-Water Dynamics in Climate Change** (cf. [OUTLOOK: RES 1](#); [Fig.3](#)), fostering unique opportunities for cross-disciplinary collaboration and solidifying its distinct research identity. The EF's contributions to our **EXCs CARE** and **REC²** echo this positive development.

Societal Change

The EF Societal Change advances TUD's distinguished legacy in social sciences and humanities research, building upon landmark DFG coordinated programmes, including CRC 537 'Institutionality and Historicity' (1997–2008), CRC 804 'Transcendence and Public Spirit' (2009–14), and CRC 1285 'Invectivity' (2017–22). Since 2019, the EF has acquired more than 20 individual DFG grants.

Our **expertise in societal change and democracy** has garnered significant recognition (total funding since 2019: €25.3M): The **Mercator Forum on Migration and Democracy** received renewal funding from the Mercator Foundation under **Hans Vorländer**, who now also chairs Germany's Expert Council on Integration and Migration. The **Hannah Arendt Institute for Totalitarianism Studies** secured funding for innovative research on climate action and far-right politics, while the Horizon Europe Project REDEMOS (PI: **Marianne**

Kneuer) advances research on Eastern European democracies. The John Dewey Research Centre for the Didactics of Democracy (PI: **Anja Besand**) pioneers innovative approaches to adult education for democracy and the Schaufler Lab, funded by the **Schaufler Foundation**, offers PhD Fellowships at the interface of technology, art, science, and society.

Further, the EF established the **TUD Disruption and Societal Change Centre (TUDiSC)** to examine societal transformations through the lens of disruption, uniting perspectives from social sciences, humanities, and engineering. TUDiSC is closely linked with the CIDS department “Speculative Transformation” (cf. [Fig.11](#)). Based on prolific research output (200 publications) and strategic appointments like **Orit Halpern** (Professor of Digital Cultures; prior Concordia University), along with notable **grant achievements** (3 DFG Research Grants; ERC Synergy Grant proposal 2nd round; EXC proposal ‘The Disruptive Condition’, a joint submission led by Leuphana University), the EF is in a strong position for upcoming funding cycles. The EF’s influence proved transformative in **co-shaping the EXC proposals REC² and BiC**, where humanities and social-science perspectives transcended traditional disciplinary boundaries to become foundational elements of these research programs, exemplifying genuine interdisciplinary integration rather than perfunctory inclusion.

(iv) Lessons Learnt

Our institutional growth in research has provided valuable strategic insights: while TUD has made commendable **progress in third-party funding**, the increase in overall DFG funding performance has not kept pace with our ambition, mostly due to a focus on BMFTR and European funding sources. This signals a need for a **refined approach to funding diversification** and to foster enhanced dialogue among University Leadership, Deans, and Faculty Members regarding DFG proposal dynamics and shifting our support mechanisms more strongly toward coordinated programmes.

Overall, the EFs have demonstrated their catalytic potential, particularly evident in their **contributions to EXC proposals and approved EXCs**. They transcended traditional disciplinary boundaries while forging robust collaborations across institutional (TUD, DDC partners) and international borders. Two EFs achieved outstanding international recognition and have been **included in respective EAs** (DIDS, Neurosciences). The others exhibited advancement toward excellence, each making notable progress in their respective areas (Mobility, Water, Society).

Observing the diverse trajectories of the Emerging Fields has illuminated a crucial insight: the imperative for **flexible support** mechanisms that respond to both developmental stages and disciplinary contexts. Moving forward our funding models, support structures, and key performance indicators (KPIs) will reflect this diversity (cf. [OUTLOOK: RES2](#)). Our journey has strengthened our conviction that excellence thrives within an inclusive research ecosystem, prompting us to balance EXU funding between **research excellence** and **pioneering initiatives** across our institution as vital incubators of future excellence (cf. [OUTLOOK: RES1](#)).

2. Teaching and Learning

In 2019, we aimed to enhance research-oriented education for talented students. We have succeeded to (i) strengthen the link between research and teaching by **integrating cutting-edge research** into our curricula and creating pathways for **talented students to engage in research activities**. Consequently, we (ii) shaped our teaching and learning strategy, emphasising **core competency development** and **project-based learning** while addressing contemporary demographic shifts in higher education.

(i) Teaching Anchored in Cutting-Edge Research and Supporting Student Research

At TUD, we interpret the Humboldtian principle of unified research and teaching as a virtuous cycle where academic learning cultivates research capabilities, while research experiences enrich the learning process. We translated this model into a comprehensive **ecosystem of research-based education** defined by three initiatives. The Teaching Synergies Programme harnesses the creative **potential of ECRs**, particularly those from our DDc partner institutions. Through professional training in interactive teaching methods and strategic pairing with TUD academics, these researchers have co-created 21 research-based courses, bringing fresh perspectives and cutting-edge research directly into the classroom.

Building on this foundation, our Teaching Excellence Tracks programme engages **senior scientists** in translating their expertise into enriched academic offerings. This initiative has yielded significant results, including the **specialised study tracks** in “Biodiversity Management” (2021) and “Forest Transformation/Governance/Dynamics” (2025).

Our EXCs have become integral to this educational enhancement, with notable examples such as **Frank Fitzek**, speaker of **EXC CeTI**, who brings cutting-edge information-science research into undergraduate electrical engineering courses. Similarly, the **EXC PoL** has advanced an international Master Programme, bridging life sciences research at the frontiers of biology, physics, and chemistry with student education.

The culmination of our research-teaching integration is the FOSTER programme, which **empowers students to conduct independent research projects**. Since 2019, this initiative has provided 316 talented students with hands-on research experience, complemented by methodological training and support for participation in national and international academic forums. The programme’s impact is particularly evident in the growing success of the annual **Student Research Exposition**, which has seen attendance surge from 70 students in 2019 to 253 in 2024. This platform not only showcases student research but also fosters peer learning and feedback, aiming to **inspire the next generation of academic researchers**.

(ii) Overall Improvements in Teaching and Learning

We have developed sTUDium 3.0, a comprehensive **teaching and learning strategy**, crafted through inclusive stakeholder engagement. This framework advances three key priorities:

What students should learn: Core competences. Beyond disciplinary expertise, we equip our graduates with essential skills for addressing 21st-century challenges such as sustainability and demographic ageing. Our competency framework, aligned with international

standards yet distinctive for TUD, will be implemented across curricula in 2026.

How our students learn: Expanding on locations and approaches. We champion problem-based learning through real-world engagement, exemplified by our students' achievements in international competitions - from Elbflorace's Formula One success to the International Genetically Engineered Machine (iGEM) Gold Medal for innovative wound treatment. Our partnerships with DDC institutions, industry, and civil society create diverse learning environments that enhance experiential education.

Studying across the life course. Responding to demographic shifts, we are expanding part-time study programmes and pioneering flexible study pathways through micro-credentials. Building on initiatives within the European University Alliance EUTOPIA and collaborations with Dresden's semiconductor industry, we plan to integrate stackable micro-credentials into our programmes by 2026 (cf. [OUTLOOK: EDU2](#)).

(iii) Lessons Learnt

While targeted initiatives have yielded valuable innovations for the research-teaching nexus, moving forward we will focus on **advancing systemic transformation** through three key approaches: First, all five funded EXCs have committed to **deeper integration of their frontier research** into Bachelor and Master coursework, catalysing innovation across our entire teaching portfolio. Second, we will be improving our administrative and legal processes for the development of Bachelor and Master programmes to create more **dynamic and research-informed curricula**. Third, we will be **engineering AI technology** to personalise learning pathways, enabling each student to optimise their academic journey toward success, while remaining mindful of the limitations and risks associated with AI (cf. [OUTLOOK: EDU1](#)).

3. Transfer

3a. Technology Transfer

In 2019, our aim was to expand our technology transfer to forge deeper industry partnerships, amplifying TUD's impact and visibility in the innovation ecosystem in Europe and beyond. Thirty-five years after reunification, none of the companies that make up more than half of Germany's total R&D spending have their corporate headquarter in the new Federal States. Despite this, TUD attracts **more industry funding** for joint research and contract work than any other university in Germany compared to its basic funding. Additionally, together with other Saxon universities, TUD leads in Germany in new business creation, making up **67% of all startups** in the region (Startup Association Report 2025). Several TUD spin-offs have grown beyond 250 employees - the EU's criteria for large companies. Combining our interdisciplinary research excellence and inventiveness with a commitment to innovation drives our success in technology transfer, evidenced by our **rank 1** in the THE Impact Ranking (SDG9: Industry, Innovation, and Infrastructure; cf. [Fig.1](#)) and our leadership in the German Economic Institute's 2024 **patent ranking**.

More specifically, our progress is demonstrated in the (i) **founding of TUD|excite**, a one-stop-shop for entrepreneurship, innovation and technology transfer. We were able to (ii)

increase the **number of start-ups**; (iii) exploit more **intellectual property (IP)** and enhance **corporate relationships**, (iv) systematically implement **training in entrepreneurship**, and (v) expand TUD technology transfer **beyond Germany**.

(i) TUD|excite: A New Hub for Technology Transfer and Innovation

Under the leadership of **Andreas Pinkwart**, Professor for Innovation and Technology Management, the founding of TUD|excite in 2023 has accelerated the commercialisation of our innovations by integrating existing and new transfer units (cf. Fig.5).

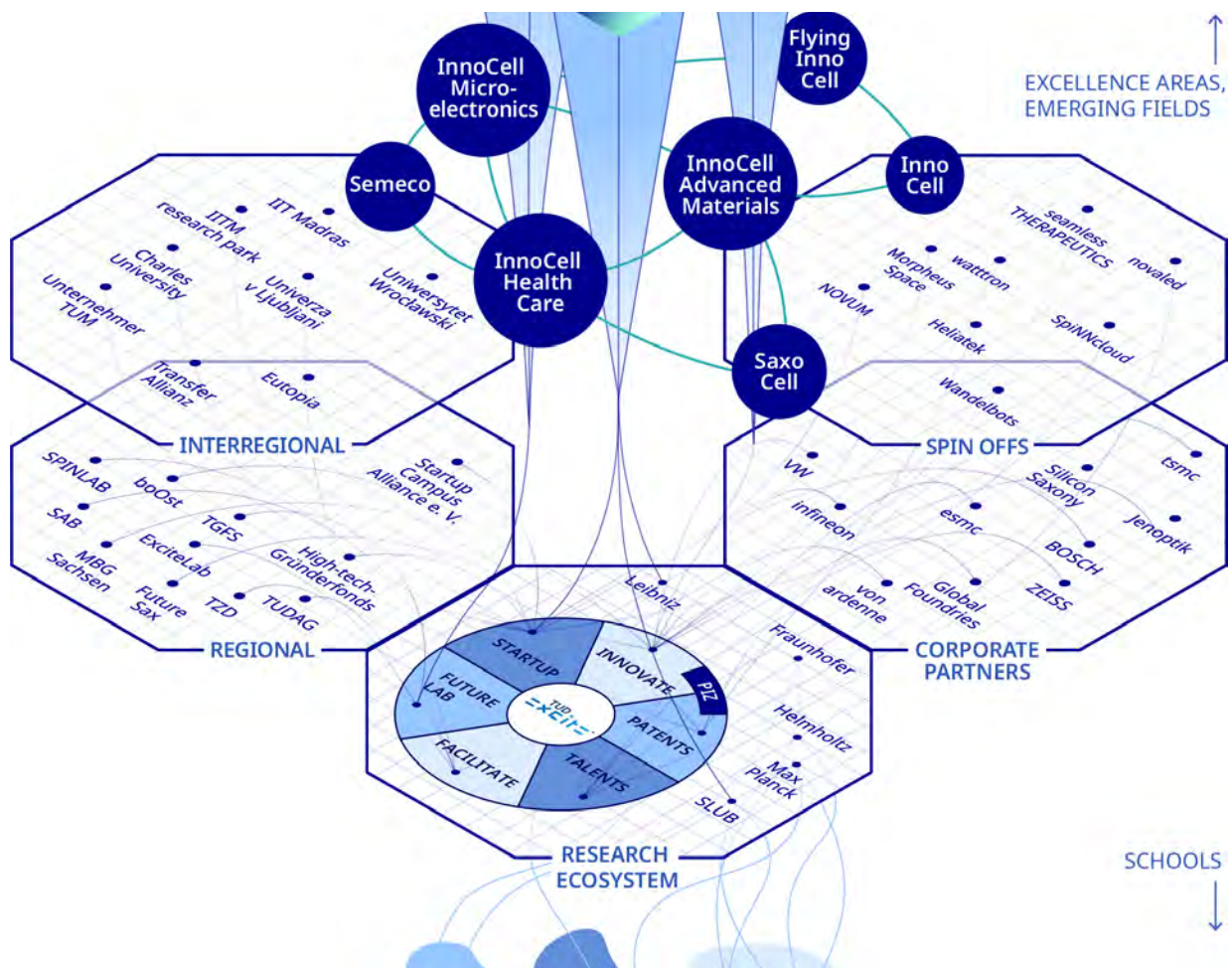


Figure 5. TUD|excite: Areas of activity; research ecosystem; regional and interregional partners; corporate partners and important spin-offs (expanded detail of Fig.3: TUD's Research Approach and Landscape).

(ii) Supporting Startups

TUD's entrepreneurial track record spans two decades, yielding over **400 startups**, including 200+ deep-tech ventures, that have generated thousands of high-calibre jobs. Our innovation portfolio features industry leaders like **novaLED** (OLED technology), **Wandelbots** (robotics), and emerging ventures such as **Seamless Therapeutics**, **SpiNNCloud** (HPC), and **Morpheus Space** (satellites). Our entrepreneurial excellence gained national recognition in 2024, with three TUD professors honoured among Germany's top 20 **university entrepreneurs** by the Stifterverband (Frank Buchholz, Karl Leo, Gerhard Fettweis).

As of 2024, our startup support ecosystem includes the **EXCITELab Accelerator**, launched in partnership with TUD incorporated company (TUDAG) and SpinLab, which has already cultivated two cohorts of international startup teams (9 teams each). This success culminated in our leadership of the **Startup Factory Business Opportunities Ost (boOst) gGmbH**, uniting ten universities from Saxony and Thuringia, to advance regional innovation (€20M federal & private funding).

(iii) Professionalising IP Management and Enhancing Corporate Relationships

TUD's **patent excellence** reflects our global innovation leadership, ranking among the top three German universities in European patent applications and securing rank **34 globally** among 1,767 institutions (German Economic Institute, 2024). We drive innovation through streamlined commercialisation pathways and targeted validation funding, bridging research outcomes to market-ready solutions.

The **excite|CLUB** established in 2024 has attracted strategic partnerships with industry leaders including **BOSCH, GlobalFoundries, infineon, esmc, ZEISS, and von ardenne** (cf. [Annex: Tab.12,13](#)). Such professional engagement of industry partners, supported by dedicated relationship managers, has built a network for TUD researchers and contributed to **four endowed professorships since 2019**. Additionally, our **excite|FUTURELAB** secured collaborative BMFTR funding for pioneering initiatives in data trust and trust incubators.

(iv) Promoting Entrepreneurial Mindset and Skills

We cultivate entrepreneurial skills and mindsets across the TUD community by uniting training on transfer and startup activities for ECRs within TUD|excite. We launched a **new certificate course on IP**, especially patent law, with a focus on research-based innovation and start-ups. In 2024, we started the **Zuse Innovators Programme Summer School** on the TUD|Campus Lusatia (cf. [RESEARCH INFRASTRUCTURE](#)) that enables teams of international doctoral students to develop business plans based on high-quality university patents. A next step will be to incorporate **entrepreneurship education** more broadly into TUD Bachelor and Master Programmes (cf. [OUTLOOK: EDU1](#)).

(v) TUD Technology Transfer Turning International

In 2019, we started expanding our activities in technology transfer beyond Germany by launching the initiative 'transCampus science to business' (**transCelerator**), together with KCL, in life sciences, medicine, and biotech. Given Germany's venture capital constraints across startup funding stages, cross-border collaboration is a strategic necessity to **expand the investment ecosystem** and enhance funding opportunities. Another strategic international partner is the **Indian Institute of Technology Madras (IITM)** (cf. [INTERNATIONALISATION; Annex: Tab.13](#)). In 2024, TUD|excite and **IITM Research Park** signed a memorandum of understanding to closely collaborate in startup activities. A joint workshop in 2025 connecting 60 entrepreneurs from both countries led to two startups – one from each country – collaborating to develop a new medical technology prototype.

(vi) Lessons Learnt

TUD stands out in Germany and beyond for turning research into real-world impact. We

identified three key goals moving forward (cf. [OUTLOOK: TRANS1](#)): (i) Innovative funding solutions are essential to bridge the critical gap in **venture capital**. (ii) Securing **sustainable funding** for technology transfer is vital, with the **Startup Factory boOst** as a cornerstone for reaching this goal. (iii) Establishing dedicated thematically-focused **innovation space and locations** will catalyse collaboration between deep-tech ventures and TUD research teams, fostering early-stage partnerships.

3b. Science Communication and Societal Responsibility

Since 2019, TUD has strengthened its societal impact through (i) enhanced science communication, (ii) youth engagement in science, (iii) public dialogue on critical issues, and (iv) advocacy for democratic values.

(i) Enhanced Science Communication

Science needs to regain or maintain public trust through conveying a better understanding of its methods and objectives. Hence, TUD has made it a cornerstone of its mission to expand its societal impact by professionalising its science communication.

TUD's science communication initiative has engaged 1,500 members through 130 workshops, pitches, and presentations aimed at **enhancing science communication skills**, especially for ECRs. This initiative culminated in the inaugural 2025 **TUD Prize for Excellent Science Communication**, awarded during the TUD Science Communication Day. Additionally, we provide robust researcher protection through **Scicomm Support**, which offers communication, psychological, and legal assistance.

Complementing these efforts, the **COSMO Science Forum** (cf. [Fig.6](#)) established in partnership with the Barkhausen Institute and DDC partner institutions at Dresden's historic Kulturpalast, serves as a permanent venue where science and society meet. Since its launch in 2022, the forum has hosted eight comprehensive exhibitions, 90 public events, and 55 interactive workshops reaching around 10,000 citizens from Dresden and beyond. Additionally, together with DDC institutions, including the Hygiene Museum, Technische Sammlun-

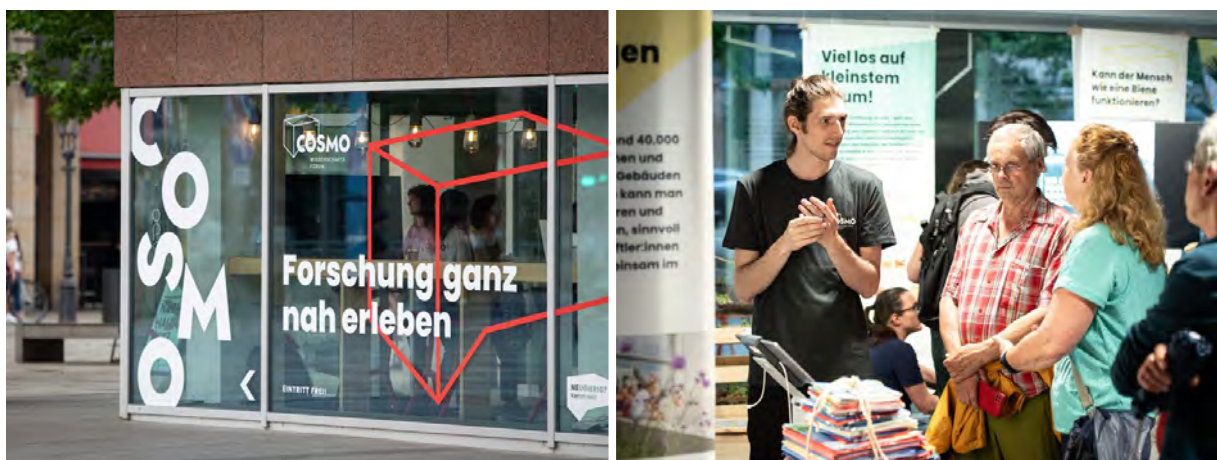


Figure 6. COSMO Science Forum: “Experience research first-hand”. Easily accessible space for TUD science communication in the city centre of Dresden that offers a platform for dialogue—where science becomes tangible, current research is explored hands-on, and critical discussion is encouraged through diverse formats.

gen Dresden, and the Saxon State and University Library Dresden we organised **exhibitions exploring frontiers in science** (2 per year) ranging from AI, to genetics and quantum materials.

(ii) Youth Engagement in Science

Through our **TUD Young Communicators** initiative, we have successfully bridged the gap between complex research and young audiences by empowering students to become science ambassadors. Since July 2021, 18 talented TUD students have created 281 videos on topics such as sustainable construction, climate change, autonomous vehicles, and HPC, reaching **780,000 YouTube viewers** and up to **600,000 views per Instagram post**, demonstrating the power of peer-to-peer science communication.

Our initiative flourishes through **strategic partnerships with local schools** (including the University School), offering students hands-on experience with cutting-edge research. **Innovative formats** include the globally successful “Kitty Q” quantum physics app (500,000+ downloads) and escape room, and CeTI’s **STEM Education programme**, which has reached 7,000 pupils through 215 modules in Saxon schools since 2019. Our **Dresden Children’s University** continues to thrive in a hybrid format, extending access across rural Saxony. TUD is successful with established formats such as the ‘**Dresden Science Night**’, which attracts more than 10,000 visitors every year to the more than 300 science events across campus.

(iii) Public Dialogue on Critical Issues

TUD’s dialogue-oriented initiatives address societal polarisation through innovative public-engagement formats. Our flagship programme ‘**TUD in Dialogue**’ has reached more than **4,000 participants** since 2019, facilitating meaningful discussions on critical issues such as climate change, migration, and diversity. The programme extends beyond urban centres into rural communities, hosting events in cultural venues, market squares, and community centres throughout Dresden and surrounding towns. This commitment to inclusive dialogue has earned recognition and additional funding support from the BMFTR.

(iv) Advocacy for Democratic Values

The pursuit of knowledge flourishes only where democracy and an open society thrive – a challenge that is particularly acute in our region. As an institution of higher learning, we cannot stand idle when **constitutional principles** and **academic freedom** are threatened. Our commitment to uphold democratic values safeguards academic independence and ensures our ability to attract global talent. As Saxony’s third-largest employer, we also bear a social responsibility to take a stance.

TUD spearheaded a **Democracy Festival** in 2024 (cf. [Fig.7](#)), uniting 70 cultural and scientific institutions and drawing 3,000 participants ahead of the European and Saxon elections. We extended our reach across Dresden, where our **community events** engaged local social housing residents on issues vital to their lives, helping bridge the perceived divide between academia and the everyday experience of people.

All of these achievements are enhanced by **expanding** our **Alumni Network** which now includes 141 Alumni Ambassadors and 18 Alumni Chapters around the world (cf. [OUTLOOK](#):



Figure 7. TUD and DDC partner institutions celebrating and defending democratic values (May 25, 2024). (a) TUD leadership heading one of the converging marches (banner says: Together for Democracy); (b) TUD Rector addressing the crowd in downtown Dresden.

TRANS2). Our graduates and former ECRs not only represent TUD’s strength in research, but also show that we prepare skilled and responsible leaders for the future.

(v) Lessons Learnt

We have learnt how investing in **young people** positively affects science communication. We want to make our science communication even **more visible** both nationally and internationally.

Engaging **sceptical audiences** demands both courage and care. To amplify our impact, we will **strengthen alliances** with other stakeholders to build a more resilient, open-minded, and engaged community.

Public engagement, particularly in regions undergoing structural transformation, demands **significant resources and expertise** to foster meaningful dialogue with communities distant from academia. To sustain and enhance our resources for public engagement, we are creating clear **indicators of impact**. These will help us better demonstrate our societal value and support third-party funding success (cf. [OUTLOOK: TRANS2](#)).

4. Research Infrastructure

In 2019, our goal was to enhance our research infrastructure in order to facilitate cutting-edge research and transform the TUD experience for the entire community. We have made progress through: (i) acquiring state-of-the-art technologies, (ii) establishing shared technology platforms and professionalising their use, and (iii) advancing research building projects across campuses. This progress is based on collaborating and pooling resources both within TUD and with our DDC partners.

(i) Acquisition of State-of-the-Art Research Technologies

Since 2019, **€73.5M** have been **invested** in **research technologies** across TUD Schools. We acquired new research technologies allowing us to advance our scientific achievements and output. Outstanding examples are a highly immersive driving simulator for scenario-based testing of automated driving functions (€6.3M), the world’s first prototype for MR-integrat-

ed Proton Therapy (€8.9M together with our DDc partner HZDR), and a Helium recovery plant for the sustainable supply of liquid helium to TUD's internal research structural units (€2.4M). Thematically, we made the most significant progress in our IT- and data infrastructures, for example, the highly energy-efficient CAPELLA supercomputer and the new supercomputer SpiNNaker 1&2 (cf. [DIGITALISATION](#)). Finally, on 8th of July 2025, as part of the two-stage **National process for prioritising new major research infrastructure** (FIS), TUD (together with RWTH Aachen, Bochum and Münster University) were selected for full proposal for the preparatory phase of the Einstein Telescope (€200M) and TUD is an associated partner in two additional successful proposals (DALI: €300M; SLICES-DE: €60M).

(ii) Advanced Technology Platforms and Shared Usage

We expanded the number and reach of our **technology platforms** by **pooling resources** for more efficient use. In addition to the four technology platforms existing in 2019 (DDc Genome Centre, Biopolis Imaging Platform, Dresden Centre for Nanoanalysis, Center for Molecular and Cellular Bioengineering [CMCB] Technology Platform), we have established five new ones for: Environmental Analytics, Chemical Analytics, Physics of Life, Neuroimaging, and Core Facilities Dresden Medicine. They are accessible for internal and external users, especially **DDc researchers**, marking a 68% increase in usage from 2018 to 2024. The management of the platforms has been facilitated by the acquisition and implementation of a **core facility management software**.

These platforms have become an **important asset** during offer negotiations with high-profile international researchers intending to join TUD, and have already contributed to **research successes** by increasing the acquisition of third-party funding and the number of outstanding publications.

(iii) New Buildings for Cutting-Edge Research and Transfer

Our commitment to advance cutting-edge research has driven substantial **infrastructure investments**, with **€363M** allocated to construction projects since 2019 and **additional €290M** earmarked for ongoing developments, including for the EXCs.

The Lehmann Office Building, the heart of our digital sciences strategy, will soon provide space for over 500 scientists and support staff in digital sciences and AI. In partnership with the DDc partner Leibniz Institute for Solid State and Materials Research, we have commissioned a state-of-the-art laboratory facility for the EXC ctd.qmat (€60M), while applying for federal funding for a new EXC PoL building (€85M). The Project House Future exemplifies our dedication to sustainable and inclusive design requiring a €78M investment. This facility will house the EXC CeTi and serve as a catalyst for interdisciplinary collaboration among TUD's academic community, industry partners, and society.

We established the **TUD|Campus Lusatia** (cf. [Fig.8](#)), which is marked by two flagship research facilities: the €74M **SML**, facilitating research in our EF Automated Mobility and our EA "Microelectronics. Information Sciences: Transforming AI" and the €54M **CircEcon building**, together with three Saxon universities, advancing our EA "Matter. Advanced Materials: Transforming Structures". The Saxon Government delegated the responsibility for these major construction projects to TUD, acknowledging our capacity for efficient project execution.

We have enhanced our **Campus Master Plan** through digital tracking solutions that optimise space utilisation for collaboration while prioritising sustainability. Integrating this with **New Work principles** like flexibility, autonomy, and purpose helps create an environment that nurtures creativity and scientific exchange across our research community.

(iv) Lessons Learnt

We gained several insights into infrastructure development that will shape our strategy. We observed that the accelerating pace of research technology advancement increasingly strains the infrastructure budgets of both universities and non-university research organisations.

As a result, we are working on **innovative financing frameworks** within the DDc Alliance (e.g., DDc Genome Centre). Furthermore, our experience led us to revise our initial 2019 approach regarding **property development authority**. Instead of pursuing completely independent property development, we have redirected our efforts toward gaining sole responsibility in overseeing the construction of selected buildings and facilitating **new models of private-public partnership** for infrastructure development in Saxony (e.g., Science Conference Centre Grillenburg; cf. [OUTLOOK: RES2](#)). Additionally, our approach to campus development has evolved, leading us to recognise that **future growth** is not merely about expanding our physical footprint through new construction. Rather, we will prioritise the efficient utilisation of existing spaces and infrastructure, embracing New Work concepts to transform our current facilities for future needs. This shift in perspective aligns with modern organisational practices and optimises resource utilisation (cf. [OUTLOOK: ENABLE2](#)).

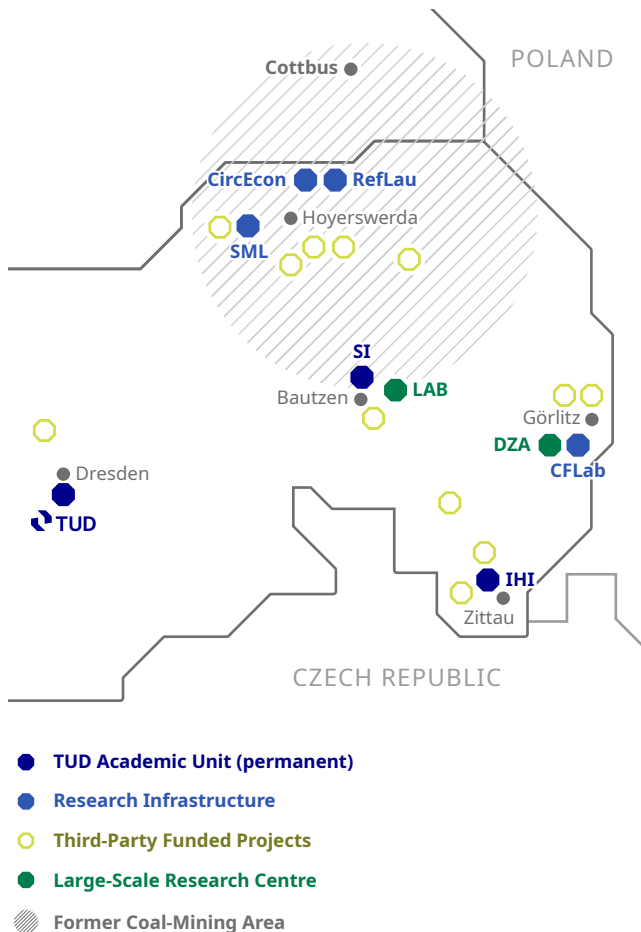


Figure 8. TUD | Campus Lusatia (ca. 90min. radius): Taking responsibility for structural transformation in a former coal mining region through research and innovation, while gaining impactful research infrastructure and the DZA, a large-scale research centre, as partner.

5. Strategic Staff Development

In 2019, we aimed to establish HR practices in line with international standards, attracting the best talent, and enabling all TUD members to develop to their full potential. We have been successful in (i) improving academic appointments; (ii) supporting early-career talent at all stages, and (iii) professionalising overall staff development.

(i) Academic Appointments: More and Faster

From 2018 to 2024, Associate and Full Professors **increased** by **12%** (548 to 614), alongside more joint appointments, EXCs, and Endowed Professorships (cf. [Annex: Tab.3.1](#)).

We introduced a **fast-track procedure** and **standardised the organisation and procedures** of the search committees, which helped expedite the compilation and legal review of the appointment dossier. Leveraging an experimentation clause in the Saxon Law for Higher Education, we were able to make **high-level strategic appointments more quickly** by targeting top scientists based on their performance and strategic fit either directly or through searches led by an international expert committee. Hence, the time from announcement to acceptance has dropped from 19 to 15 months (since 11.2019), making the process **21% faster**.

We established **active recruiting procedures**, aimed at international talents and outstanding female scientists. Consequently, we **increased** the overall proportion of **female professors** from 17.9% (n=98) in 2018 to 26.7% (n=164) in 2024 (cf. [Annex: Tab.3.1](#); [Fig.9](#)). This increase was bolstered by the creation of a comprehensive **Dual Career** and **Welcome Service**. It includes finding positions for partners, securing high-quality childcare and helping newcomers in making Dresden their home. More than 50% of all new appointees and 60% of internationals used this service (cf. [INTERNATIONALISATION](#)).

(ii) Empowering ECRs at all Stages

ECRs are highly productive, yet the German academic system constrains their autonomy and potential. Since 2018, we have increased the **number of Junior Professorships by 55%** and those with **tenure track by 100%** (cf. [Annex: Tab.3.1](#)). Using EXU funding, we have implemented additional successful measures to counteract the constraints of the current system:

- (1) The **TUD Young Investigator** status for independent research group leaders (48 since 2019) allows them to act as primary supervisors for doctoral candidates and awards faculty rights. To date, 80% of awardees have achieved tenured professorships.
- (2) The **DDc Research Group Leader** award for outstanding independent ECRs (8 since 2019) is a matching-funds model with our DDc partner institutions, enabling highly talented researchers to autonomously advance their research programmes.
- (3) The **Postdoc Centre** (together with DDc partner HZDR), established within the **TUD|Graduate Academy** (TUD|GA) represents a strategic enhancement of our post-PhD career support infrastructure. We have developed five specialised professional **qualification tracks**, each offering mentoring and targeted qualification programmes spanning scientific research, university teaching, scientific research infrastructure coordination, scientific research management, and industry engagement.

TUD|GA serves as a cornerstone of academic development, providing **comprehensive support** to over 6,000 ECRs, including both doctoral candidates and postdoctoral scholars. It offers individual coaching, professional development initiatives, funding opportunities, and mental-health support. Further, we have instituted robust **quality assurance** measures that exceed DFG standards, implementing enhanced doctoral supervision protocols that encom-

pass dual mentorship, formalised agreements, and systematic annual evaluations to ensure academic excellence, and **mutual accountability**.

(iii) Professionalising Staff Development

Consistent with our third-party-funding success, between 2018 and 2024, the number of **scientific staff** (FTE) grew by 7.5% (3,970 to 4,267), while **support staff** increased by 10.2% (2,309 to 2,544) (cf. [Annex: Tab.6](#)).

Demonstrating our commitment to attracting and nurturing top talent, we have created a **Talent Development Unit** within our HR Division, enabling comprehensive employee lifecycle management, employer branding, and seamless coordination between HR administration and development initiatives. Leading diverse teams in administration and research is complex, therefore we established **TUD Leadership Guidelines** through inclusive stakeholder collaboration. These guidelines, adopted in 2023, embody our core social values – integrity, respect, openness, trust, and transparency – while setting clear standards for leadership practices. As we are implementing these guidelines, **leadership training** is tailored to individual needs. To address growing regulatory demands and ensure compliance, we have formalised previously informal academic leadership practices through clear, standardised frameworks that promote **equitable treatment and legal transparency** across TUD.

(iv) Lessons Learnt

Our strategic initiatives in academic recruitment have delivered significant value, but sustainable progress demands targeted optimisation. In order to continue to improve academic recruitment, we have identified three priorities: completing the **digitalisation of appointment procedures** to boost efficiency, implementing a **competency-based recruitment framework** to ensure quality and attract the right talent, and increasing the **use of the tenure-track option** to increase our competitiveness. These refinements will enhance resource allocation and reinforce our institutional excellence in academic appointments.

While our **ECR support programmes** have achieved remarkable success, **fiscal sustainability** calls for a new approach. We will develop a **funding model** that involves stakeholders to ensure the longevity of our differentiated support system without compromising its quality or reach.

The transformation of our HR practices has underscored the pivotal role of **organisational culture** in driving change. Success depends on cultivating a value-driven environment of continuous learning, reinforced by consistent leadership. To address organisational barriers, we introduce flexible, **workload-conscious professional development** opportunities. Additionally, the rise of hybrid academic and non-academic careers calls for a more **integrated approach to staff development** (cf. [OUTLOOK: ENABLE1](#)).

6. Equal Opportunity & Diversity

In 2019, we aimed to promote equal opportunity and diversity within a **university culture** of respect. Four major achievements reflect our progress: (i) establishing the Office of Vice President for University Culture, (ii) implementing innovative tools for women's equal op-

portunity, (iii) promoting diversity through various instruments, and (iv) fostering transparency and active participation.

(i) Establishing the Office of Vice President for University Culture

As the first German university to establish a Vice Presidency for University Culture in 2020, TUD affirmed its commitment to fostering an open, diverse, and vibrant community for all members. This strategic office has advanced key issues including ecological sustainability, compliance, inclusion, and societal responsibility. Hence, TUD has launched a comprehensive **sustainability strategy**, established a **Green Office**, and developed a robust **compliance management system** to support good scientific practice, anti-discrimination, and anti-corruption efforts.

(ii) Innovative Tools to Promote Equal Opportunity for Women

Our commitment to equal opportunities for women has yielded significant results through **targeted fellowship programmes** supporting female researchers (Eleonore Trefftz Professorships: 3 p.a., Maria Reiche Programme: 20 p.a.) and expanded **leadership support initiatives** (Women Leading in Academia: 6 p.a., Women@DDc network of ca. 500 members).

Our measures have driven a **steady rise** in female professors and scientific staff (cf. Fig.9; Annex: Tab.3.1 & 6), particularly within our leadership structure. TUD now features a gender-balanced University Executive Board (EB), increasing from 0 of 5 in 2019 to 4 of 8 since 2020. Additionally, the number of female Deans has grown remarkably, from 1 of 17 (6%) in 2019 to 7 (41%) in 2025.

(iii) Promoting Diversity Through Diverse Instruments

Building on our Diversity Strategy 2030, TUD has successfully completed the ‘**Shaping Diversity**’ audit. We value diversity as a key scientific and organisational asset and promote inclusion throughout university life. Our TUD Leadership Guidelines make inclusion and diversity management a **core leadership responsibility** (cf. STAFF DEVELOPMENT).

Since 2019, we have strengthened support for underrepresented groups through, for instance, anti-discrimination workshops that engage approximately 1,000 participants annually. Developed collaboratively, our **TUD Inclusion Action Plan 2.0** sets out 35 measures to enhance accessibility and inclusion. Additionally, our **Incentive Fund for Equality and Diversity** has funded 33 bottom-up proposals since 2021.

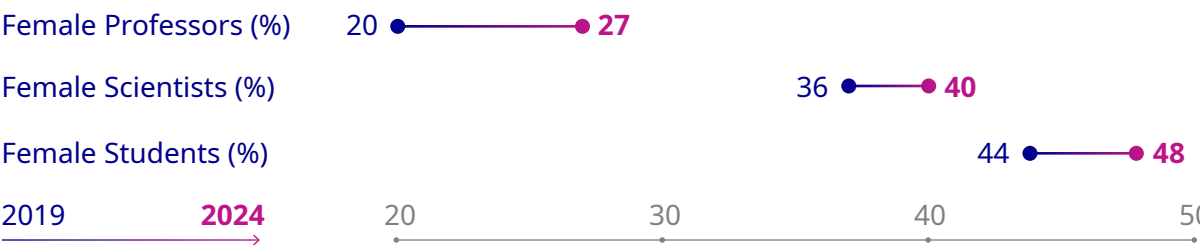


Figure 9. TUD: More female than ever (%).

(iv) Fostering Transparency and Active Participation

Creating effective internal communication tools to enhance transparency is vital for advancing university culture. We distribute **University EB meeting minutes** to second and third-tier leadership, including student government, making it one of the most frequently read TUD documents. The latest employee survey indicates that TUD members feel better informed than in previous years, particularly regarding decisions made by the EB. Our monthly newsletter **UpTUDate** reaches 40,000 TUD members.

The online platform **TUD Impulse** allows community members to submit and discuss suggestions for university improvement. In its first year 77 ideas were submitted and four implemented, engaging more than 1,500 community members. Additionally, the **TUD University Forum** serves as a venue for engaging in discussions on controversial topics, such as climate change or abuse of power, promoting informed and balanced dialogue and encouraging diverse perspectives.

(v) Lessons Learnt

We learnt that successful cultural transformation requires persistent engagement across all university constituencies. While we see meaningful progress, we recognise that transforming institutional culture is always a work in progress. Sustained commitment, clear objectives, transparent communication, and accessible formats have proven essential.

While we have advanced significantly in women's equal opportunities, strengthening support for **other underrepresented groups** remains a priority. We have learnt that embedding diversity as a **leadership responsibility** at all levels is crucial, with **middle management** playing a vital role. Moving forward, we recognise the need for more systematic **programme evaluation** to drive improvements and attract additional external funding. Drawing on these learnings and our ongoing commitment to cultural transformation, TUD is building **resilience and agility to thrive in a changing environment**.

7. Internationalisation

In 2019, we set out to strategically intensify internationalisation at home and global collaborations. We have successfully achieved (i) an increase in the recruitment of international researchers and students, (ii) improved integration of our international staff and students, and (iii) a strategic expansion of our international collaborations.

(i) Attracting International Researchers and Students

We successfully attracted **more international talent** through enhanced international visibility of our EXCs, actively recruiting exceptional researchers, such as the **Humboldt Professor Miki Ebisuya-Matsuda** (prior EMBL Barcelona) or ERC grantee and **EXC Speaker Yana Vaynzof** (prior Heidelberg University and Cambridge University) and by refining our appointment process. Additionally, we have increased the international diversity of our student body through **targeted marketing** efforts and an increased number of English-taught programmes. [Figure 10](#) summarises key indicators of progress.

Our **TUD Fellows Programme** has emerged as a catalyst for international collaboration, welcoming **89 distinguished researchers** since 2019 from leading global institutions for re-

search stays ranging from two weeks to six months. The programme’s impact is reflected in **scholarly output**, 55 joint publications, 10 successful conferences with two more in development, seven submitted grant applications with another in preparation, two summer schools and innovative outreach initiatives, and support with strategic TUD development projects (e.g., advanced technology in education).

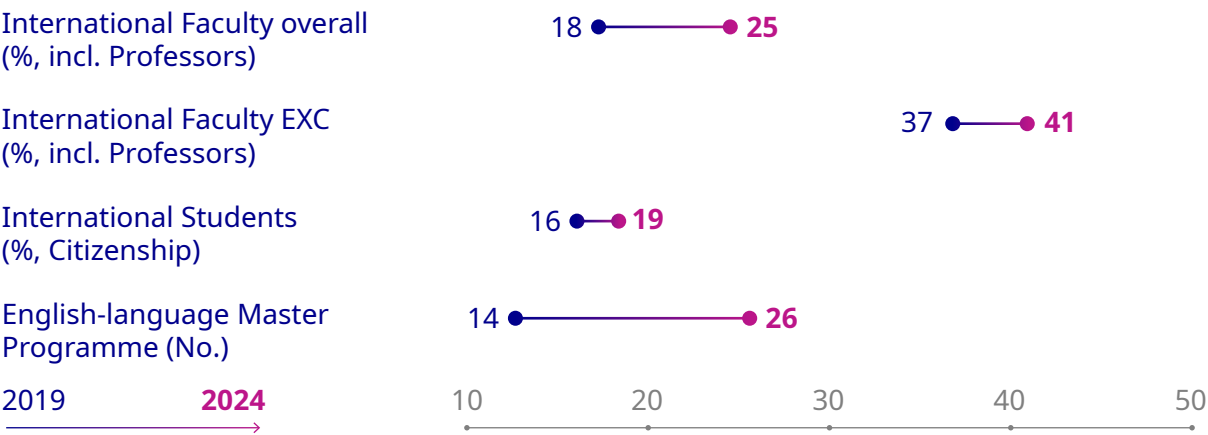


Figure 10. TUD: More international than ever.

(ii) Integrating International Staff and Students into the TUD Community

The **DDc Welcome Centre** serves as the cornerstone of our comprehensive international support system for newly arrived internationals, providing essential services ranging from visa processing and healthcare access to housing and childcare arrangements for accompanying family members (**281 internationals** were served in **2024**). Our integration strategy further includes **dual-career support**, **language courses**, and customised buddy and **mentoring** programmes.

Our commitment to inclusivity is reflected in our **bilingual campus environment**, with all official communications available in German and English. In 2024, 456 members overall participated in English and German language courses, intercultural and English workshops, as well as individual coachings and language learning apps. The newly launched **Global Citizen Certificate** further enriches our international landscape, offering students opportunities to develop global perspectives and competencies through lectures and public engagement activities, advancing our **internationalisation@home strategy**.

(iii) Strategically Expanding and Strengthening International Collaborations

transCampus: A Unique Partnership Model

Our transCampus model is unique worldwide. It addresses research, teaching, and transfer and entails **infrastructure sharing**, **joint professorships**, and **collaborative third-party funding**. The **transCampus with KCL**, established in 2013, has expanded from medicine and life sciences to engineering, generating 100 joint peer-reviewed publications annually, over 200 PhDs, and €12M in third-party funding (2019-25). Core initiatives include two IRTGs and two CRCs.

Our latest transCampus partnership is with the IITM, India’s premier science and tech-

nology institution. This strategic collaboration leverages India's young demographic and renowned academic excellence, particularly in technological innovation and research. Anchored by a €2.8M Research Centre on Global Water and Climate Adaptation, the **transCampus TUD-IITM** has launched the first Indian-German Joint Master's Programme and has also fostered synergies in transport sciences, microelectronics, medicine, and technology transfer (cf. [TECHNOLOGY TRANSFER](#)). Efforts to build a **transCampus with University Zurich and ETH Zurich** are in progress and have already led to DFG funding in medical sciences (IRTG 3019).

Science Liaison Offices: Strategic Engagement

Our Science Liaison Offices (SLOs), operated under the auspices of the Saxon Ministry of Science, enhance our international engagement through research collaboration, talent acquisition, and technology transfer. The **Taipei SLO**, established in 2023, has cultivated partnerships with prominent Taiwanese institutions and coordinates the **Semiconductor Talent Incubation Programme Taiwan (STIPT)**. The programme's outstanding success has prompted the BMFTR to appoint **TUD as the national coordinator** for expanding STIPT across German universities.

Our **Chennai SLO** (Tamil Nadu, India), which opened in 2024, further strengthens our already robust network in India (transCampus TUD-IITM, HERITAGE network). We have also forged new collaborations with **other premier IITs**, including those in Dhanbad, Mandi, Delhi, and Roorkee.

European Integration and Regional Leadership

Dresden's strategic location and current geopolitical dynamics have intensified our European focus. Our membership in the **EUTOPIA Alliance** (EU-funded: €4.4M, TUD share €1.2M; cf. [Annex: Tab.13](#)) not only facilitates innovative formats for research-based teaching and learning, exemplified by EUTOPIA's 'Connected Communities' and **European micro-certificates**, but also **eases student and faculty mobility**, and expands available **research expertise** for European grant applications.

Our position in the **Polish-Czech-German tristate area** stimulates regional initiatives, including **CASUS** (cf. [Fig.4](#)), an HZDR research institute focused on complex-systems science located at the Polish border and involving the University of Wrocław, the **Building Bridges Conference**, which link scientists and policy makers from the three countries, and the **Central European Campus Neisse/Nisa**, associated with the **TUD|International Institute Zittau (IHI)**. Since 2012, IHI is a Central Academic Unit of TUD and recently has become a focal point of our **TUD|Campus Lusatia**. This strategic positioning has fostered extensive academic partnerships, resulting in **double-degree programmes** with leading institutions in **Wrocław, Liberec, and Pardubice**, while facilitating shared courses with **Charles University Prague** and collaborations with the **Czech Academy of Science** in doctoral training in the life sciences (DIGS-ILS).

(iv) Lessons Learnt

The cross-cutting nature of internationalisation necessitates seamless integration across

all major fields of TUD's activities while navigating the emerging political complexities of **export control** and **global engagement**. Our forward strategy centres on four key areas: first, the strategic deployment of the **TUD Fellows Programme** to enhance our EFs and to bridge between disciplines; second, improved **student recruitment** through EXCs and local onboarding programmes, alongside better pathways to academic success and integration into the German labour market; third, the implementation of **AI-based English interfaces** to support administrative staff and strengthen international representation in university governance (cf. [DIGITALISATION](#)); and fourth, the establishment of a third **transCampus**, which underscores the importance of institutional flexibility in international partnerships and informs our approach to future collaborations (cf. [OUTLOOK: DRIVE3](#)).

8. Governance

In 2019, we aimed to further improve our capacity for effective decision-making and institutional renewal. Our achievements are: (i) more effective central and decentral governance, (ii) leveraging diverse perspectives to improve capacity for continuous strategic renewal, and (iii) improved academic data management and quality control.

(i) Effective Centralised and Decentralised Governance

The evolution of TUD's governance structure underscores our commitment to balanced institutional development and strategic innovation. The creation of five **TUD Schools** in 2017 has significantly enhanced collaboration among Faculties and streamlined communication with TUD's Executive Board, as confirmed by an external evaluation in 2022 (see [Annex: Tab.17](#)). This restructuring has fostered administrative synergies, raised the visibility of the Schools, and led to ongoing efforts to clarify responsibilities among Schools, Faculties, and Central Administration.

In **managing** our **EXCs**, we have adopted a strategic approach that prioritises scientific agility over institutionalisation. As a rule, each EXC should operate for a maximum of 14 years, after which its professorships are reintegrated into the Faculties and its research infrastructure is converted into technology platforms or TUD|Science and Resource Centres. This model, currently being implemented with two former EXCs, CRTD and the Centre for Advancing Electronics Dresden (cfaed), promotes the regrouping of interdisciplinary collaboration within faculties to drive innovation and address new scientific frontiers.

To address increasingly complex institutional challenges and adapt to changes in our external environment, we strategically **restructured the TUD Executive Board (EB)** in 2020. This reconfiguration expanded its focus to include key areas such as **digitalisation, internationalisation, technology transfer**, and **university culture**, alongside traditional academic portfolios like research and academic affairs. To ensure coherence in strategic development, we have fostered transparency and task-related collaboration within the EB. Additionally, the responsibility for **University Development**, previously managed as a distinct office within the EB, has been integrated into the **Rector's portfolio** (cf. [Annex: Tab.17](#)).

To align with the new composition of the EB, we established **new Divisions** for Communication, University Culture, Digitalisation, and University Development through reorganising existing ones. Simultaneously, we implemented the principle of task-related responsibility

within the administration, which transcends Division-related responsibility and traditional administrative hierarchies within Divisions. This approach has enhanced effectiveness while carefully monitoring staff workloads.

Establishing the **Division for University Development** has significantly strengthened our **capacity for integrated strategic action**. As TUD develops domain-specific strategies, such as those for sustainability, equal opportunities, and teaching and learning, the Division for University Development ensures cross-links and coherence, addressing a weakness identified in 2019. It serves as an integrating unit for strategic initiatives, institutional advancement, and transformation monitoring, reducing fragmentation and friction. Moreover, it works to **make the most of EXU funding to bring positive change** to all of TUD.

(ii) Leveraging Diverse Perspectives to Improve Capacity for Continuous Strategic Renewal

In line with TUD's emphasis on **diversity of perspective as a resource** (cf. [EQUAL OPPORTUNITY](#)), we have restructured decision-making bodies and processes to integrate a broader range of viewpoints (cf. [Annex: Tab.18](#))

Annual **leadership retreats** are organised to foster strategic dialogue between the EB and key stakeholders, including representatives from the **Senate, Schools, Faculties, Central Academic Units, and Divisions of the Central Administration**. These retreats, refined through joint preparation, and systematic evaluation, serve as dynamic forums for rigorous analysis, open discourse, and collaborative insights that shape institutional direction.

To further harness the knowledge of the **TUD community**, we organised seven **Future Labs**, open to all members, where the EB presented strategic goals for feedback and new ideas. These participatory formats have helped to design domain-specific strategies, such as those for sustainability, equal opportunities, and teaching and learning. Since 2019, **over 600 TUD members** have contributed to **shaping our institutional strategy**.

Recognising the value of external perspectives, we have also developed tools to analyse the global higher education landscape. **Systematic benchmarking** against leading international universities has become a cornerstone of our strategic process, helping us refine our competitive profile and align with emerging standards of excellence, regarding, for instance, HR practices, innovative, co-creative education, alumni work, or fundraising. Partnerships within the EUTOPIA Alliance (cf. [INTERNATIONALISATION](#)) and with institutions like RWTH Aachen, Technical University of Munich, and ETH Zurich have been particularly valuable for **peer learning** and strategic positioning.

The TUD **International Strategy Council** (ISC; cf. [Annex: Tab.18](#)), established in 2019 and composed of experienced international science managers, serves as a crucial instrument for **strategic guidance** and ongoing evaluation. In January 2024, ISC members played a pivotal role in **our self-initiated EXU Midterm Review**, offering essential insights into our pursuit of international excellence and impact. The evaluation identified key strategic priorities for achieving parity with leading global institutions by refining our strategic vision and its articulation, harmonising our diverse portfolio of initiatives under this unified vision, and strengthening our commitment to systematic strategic monitoring.

(iii) Improved Academic Data Management and Quality Control

The development of our **institutional monitoring and evaluation system** underscores our dedication to data-informed excellence and strategic growth. Since its launch in 2017/18, with an initial emphasis on research excellence, our **Faculties Evaluation System** has evolved into a comprehensive institutional framework that assesses the full breadth of academic activities – research, teaching, and transfer. Between 2019 and 2025, the systematic implementation of **external peer reviews** across nine faculties has resulted in strategic action plans, collaboratively crafted by faculty leadership and councils together with the EB. These plans, supported by annual monitoring processes, ensure **alignment with TUD's institutional strategy** while strengthening faculty-level strategic capabilities and enhancing the coordination of third-party funded research programmes and excellence projects (cf. [RESEARCH](#)).

Recognising that deep institutional self-knowledge is the foundation of strategic excellence, we significantly advanced our **data management and analytics capabilities**. A data-quality management framework has been established, integrating clear data stewardship with dynamic **performance visualisation tools** to enable evidence-informed decision-making across TUD. By strengthening feedback mechanisms between organisational levels and building strategic capacity within Faculties, we have cultivated a culture of trust in our data governance, laying a robust foundation for sustainable academic analytics (cf. [Annex: Tab.18](#)).

In this context, the 2022 implementation of our **Research Information System** marks a significant advancement in our ability to monitor and evaluate research performance across TUD. Since 2019, it has grown to over **1,500 researcher profiles** featuring over **100,000 achievements** (mostly publications and grants) that are now publicly visible through the **TUD research portal**. This standardised information lays the groundwork for decentralised incentive systems, including target agreements and performance-based fund allocation, while supporting our strategic goals in internationalisation, equal opportunities, and sustainability. Looking ahead, we aim to integrate this system within our overall academic data management by 2030, promoting comprehensive academic analytics and continuous quality improvement.

(iv) Lessons Learnt

Our Excellence title has both empowered and obligated us to strengthen our strategic capabilities through rigorous analysis of external trends and internal developments. This process has yielded key insights: First, effective institutional transformation requires an integrated and systematic approach. To this end, all EXU measures must align with TUD's overarching strategic objectives to ensure cohesive institutional progress. As a catalyst, we plan to establish a dedicated **transformation management unit** in 2027 to foster cross-structural collaboration. Second, maximising institutional impact necessitates focusing on strategically significant initiatives rather than diluting resources across isolated, smaller projects. This **prioritisation** must be informed by **strategic foresight, robust data analytics**, and thorough **cost-benefit analysis**. Third, we have recognised that the success of strategic development depends on **cultivating change-readiness at all levels** of TUD.

Our institutional strategy for the period of 2027 to 2033 will address both the organisational and cultural dimensions of change to drive holistic institutional evolution (cf. [OUTLOOK: ENABLE2+3](#)).

9. Digitalisation

In 2019, we identified digitalisation as a key driver for overall institutional innovation and for facilitating research, teaching, transfer, and administration. Our key achievements are (i) implementing a new digitalisation governance and (ii) establishing CIDS as a science and resource centre, enhancing our digital capabilities and support structures for all members and disciplines.

(i) Implementing a New Digitalisation Governance

In 2020, we introduced the Chief Digitalisation and Information Management Officer (**CDIO**) to the University EB. The CDIO **reformed digitalisation governance** by creating a CDIO Strategy Council, an IT Coordination Team, and a Steering Group Digital Administration, ensuring coherence in strategies, effective digital services, and prioritisation of transformation projects.

Given the steadily increasing complexity in IT, we responded by integrating **central and de-centralised IT**, creating an **TUD|IT asset map** and harmonising IT resources hence promoting efficiency and building resilience. Thus, we facilitated university-wide **security measures** and **sharing of IT resources** across Schools and Faculties. Additionally, standardised digital workstations were rolled out, streamlining replacements, reducing administrative effort, improving security, and cutting costs. The way we integrated central and decentralised IT support has served as a role model for other universities.

We focused on improving digital support for administrative processes and introducing new management principles. We integrated TUD's digitalisation activities under the umbrella of a **digitalisation multi-project management**. New **tandem leads** combine domain and IT expertise. We have increased **transparency** in digitalisation through participation of users, holistic monitoring, and realistic cost/feasibility assessments. **Notable projects** accomplished since 2019 include digital workflows for hiring student assistants, vacation requests, and invoice processing. Enabling paperless workflows has reduced processing times from days to hours, and facilitated remote work.

(ii) Establishing the Science and Resource Centre for Interdisciplinary Digital Sciences (CIDS)

In 2021, we established CIDS, a unique **Science and Resource Centre**, previously referred to as the 'Lehmann Centre' in the 2019 proposal, that integrates digitalisation research with top-ranked computing and AI infrastructures, alongside essential digitalisation services and support units. For the first time in TUD's history we have unified research support and related sciences within a single centre, allowing operations and research to inspire one another for enhanced innovation and service quality.

CIDS has become a key player in the fields of **research data infrastructures, HPC and AI** (cf. [RESEARCH](#)). In HPC, we rank among Germany's top five universities, thanks to the energy-efficient

CAPELLA supercomputer (3rd in Germany, supported by the DZA), the SpiNNaker1&2 supercomputers, and a pioneering quantum computing lab (total invest: €60M; cf. [RESEARCH INFRASTRUCTURE](#)).

CIDS serves as a basis for several high-profile interdisciplinary research projects and collaborations. It has proven to be attractive for talent in the competitive field of digital sciences and currently oversees approximately 500 science and science-support staff in digital sciences and AI, with expectations to grow to 600 by the end of 2026, including at least 25 professors from all TUD Schools who hold secondary affiliations with CIDS.

Figure 11 shows that CIDS to date comprises eight departments, five of which focus on combining digital-science research with services: (i) ‘Scalable Data Analytics and Artificial Intelligence’ (ScaDS.AI Dresden/Leipzig; 2019-24: €29M for TUD), one of six **National AI Competence Centres**; (ii) the department ‘Information Services and High-Performance

Computing’ as one of the nine sites of the **National High-Performance Computing (NHR)** network.

This portfolio is complemented (iii) by a department for ‘Research Data Analytics’ forming the core of our significant involvement in the development of the **National Research Data Infrastructure** and the **National Data Competence Centre Come2Data** (cf. [RESEARCH](#)). (iv) The department ‘Computational Modelling and Simulation’ (CoMoS) completes the CIDS data-analytics portfolio. CoMoS is a think tank, led by **Ivo Sbalzarini**, Professor of Scientific Computing for Systems Biology and Senior Research Group Leader at MPI CBG, promoting novel and unconventional research approaches to **modelling and simulation** that integrate AI/machine learning with numerical analysis methods and closely cooperates with our DDc partners. (v) Connecting with academic affairs, the Centre Open Digital Innovation and Participation (CODIP) translates digital methods into **teaching tools** that are easily accessible. Furthermore, focusing on service only, CIDS just launched the TUD|Support Centre Digitalisation

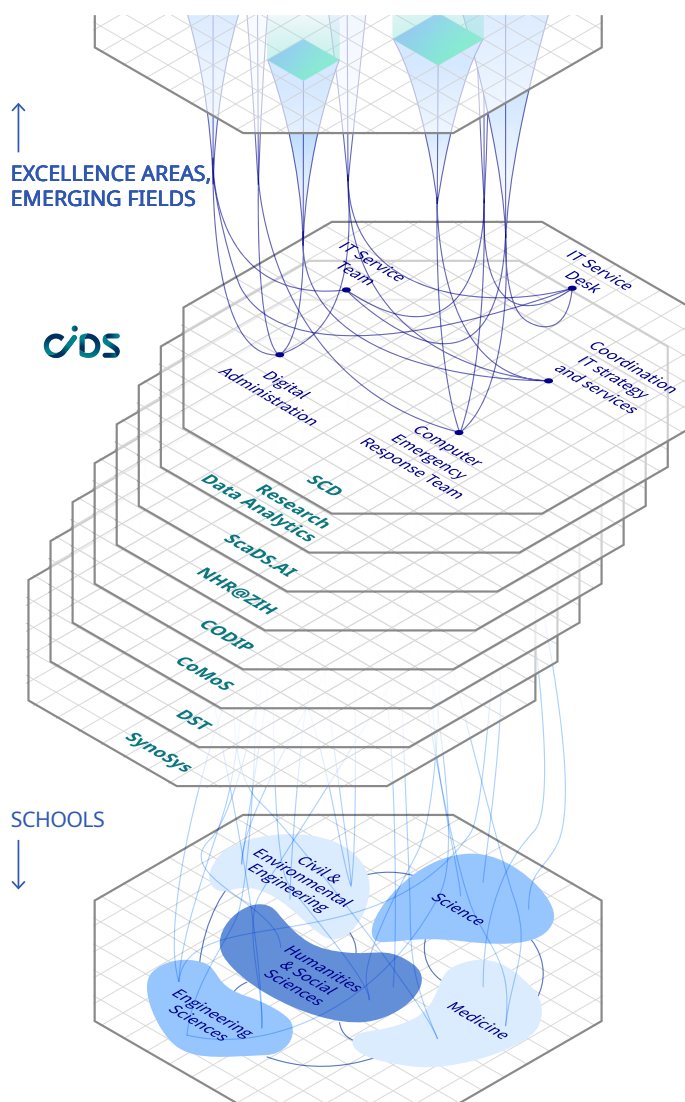


Figure 11. The Centre for Interdisciplinary Digitalisation CIDS comprises 8 departments: 5 combine science and service (Research Data Analytics, ScaDS.AI, NHR@ZIH, CODIP, CoMoS), 2 are research-focused (SynoSys, DST; cf. [RESEARCH: DATA-INTENSIVE SCIENCE](#)) and a one-stop service unit (SCD) (enlarged detail of Fig.3).

(SCD) as a **one-stop-shop for digital support**, cybersecurity, and digitalisation of administrative processes.

(iii) Lessons Learnt

CIDS's multidisciplinary approach uniquely integrates technical, systemic, and social aspects of digitalisation, positioning **TUD as a leader in AI and HPC**. Hence, CIDS may serve as a flexible framework for collaboration and a **potential blueprint** for the establishment of **other Science and Resource Centres** at TUD (cf. [OUTLOOK: RES2](#)).

To successfully implement digitalisation-driven change, all stakeholders must be involved. Domain experts need adequate **time for co-design**, and assume **shared project ownership** with digitalisation experts (i.e., **tandem lead**). Given **rising cyber threats**, strengthening TUD's digital security remains central to our strategy. We need to reach for the **next level of operations** by **saving time and human-resource** investments while increasing quality and performance through including AI services in research, teaching, transfer and administration (cf. [OUTLOOK: DRIVE1+2](#)).

Part 2: Outlook

A. Advancing Excellence in Challenging Times: Vision, Mission, and Identity

Not least due to EXU funding, **our journey is one of strategic advancement**. We joined the ranks of Germany's top five universities and top 10% globally (cf. Fig.1). We have expanded our global impact, benchmarked against international standards and forged worldwide strategic partnerships. Through collaborations across scientific, political, economic, and cultural spheres, we have established a transformative regional role.

TUD's development demonstrates our strategic agility and resilience and how we affect both academia and society. These strengths are especially valuable today, amid renewed military conflict in Europe, economic stagnation, and widespread attacks on democracy – all contributing to uncertainty. Hence, our strategic plan 2027-33 is vision-driven, yet contextualised.

We envision **TUD as a resilient research university of excellence, pioneering transformative solutions for the 21st century, where global impact meets regional leadership**.

We have made it our **Mission** to expand the boundaries of fundamental research, educate responsible leaders for tomorrow, develop purposeful breakthrough technologies, and engage in societal dialogue in divisive times. We will strengthen our institutional resilience and generate transformative solutions for selected global challenges (i.e., climate crisis, resource scarcity, demographic ageing, and the jeopardy of democracy).

To reflect our history and advance our Mission, we engaged the TUD community in a comprehensive process to define our **Institutional Identity**: TUD is the **Collaborative University: inventive. transformative. engaged**.

Collaboration is our core: TUD's advancement reflects its collaborative power. Collaboration goes beyond cooperation, it means working together towards common goals, sharing responsibilities, knowledge, and resources. We collaborate across disciplines, with industry, societal partners, and globally.

We are **inventive**: We are known for inventions such as the fundamental improvement of OLEDs or developing the knowledge base of Wikipedia through Wikidata. We are creating new ideas and solutions that show technological progress and non-obvious problem-solving for societal and epistemic benefit, as well as in teaching, talent attraction, societal engagement, and organisational operations.

We are **transformative**: We actively shape transformations in science, technology, and society by shifting perspectives, updating systems, and improving practices. We do so responsibly: we consider unintended and long-term consequences.

We are **engaged**: We are involved and deeply committed to address the most pressing issues of our time. We aim to make a difference both locally and globally. Our engagement also implies to show genuine commitment to our work, institution, and each other.

B. Strategic Areas and Approach

Our new strategy focuses on pushing **transformational innovation** and building organisational **resilience** (cf. Fig.12). Our strategic approach focuses on **three core performance areas**: research, education, and transfer. Two **driving** forces will boost innovation: Internationalisation increases diversity of perspectives and enhances resilience by enlarging our scope of action. AI-enriched digitalisation will transform how we operate and collaborate in research, teaching, transfer, and administration.

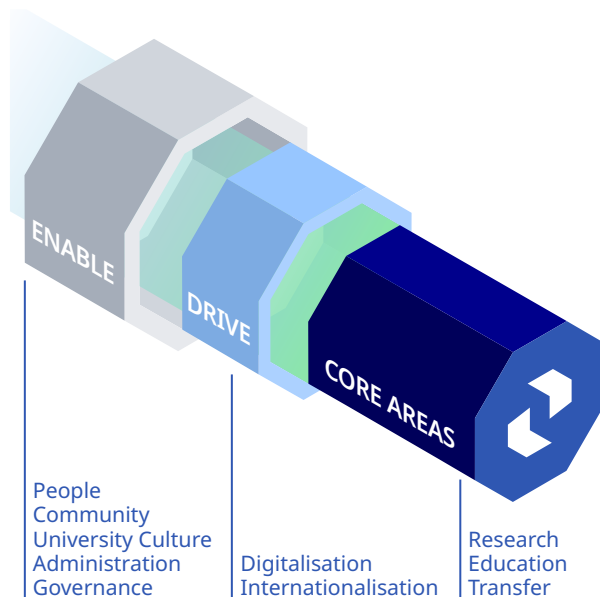


Figure 12. TUD's strategic approach: Research, education, and transfer are our core performance areas. These are enabled by resources such as university culture, people, administration, and governance, while internationalisation and digitalisation are important drivers of innovation.

We identified three **enabling resources** as indispensable for the envisioned organisational transformations allowing us to leverage opportunities and master challenges: (1) Ensure effective governance and administration that enables us to combine strategic foresight with sustainable strategic decisions and optimised processes. (2) Create a resilient TUD culture and community. (3) Empower every TUD member to realize their full potential.

Both drivers and enabling resources are not ends in themselves, but are measured against their impact on our three core areas of activity. The following plan is derived from this strategy and describes how EXU funding will allow us to pursue the activities we deem critical to expand our institutional excellence.

C. Plan of Action and Goals

1. Core Areas of Performance

RESEARCH

RES 1: World-Class Research

We will strengthen our Excellence Areas through our EXCs and new EFs that are science-driven and address selected global challenges. Research KPIs are used to assess progress.

To reach this goal, our research approach will focus on three, closely connected **EAs** (cf. Fig.3), enhancing our institutional profile and visibility. Five EXCs reflect world-class science and position us well to develop transformative solutions for chosen global challenges. Figure 13 illustrates their strong interdisciplinary nature.

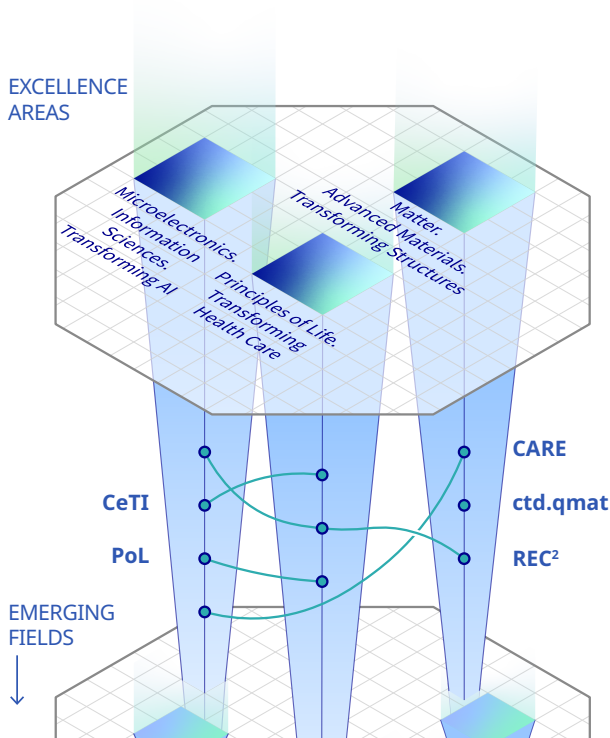


Figure 13. Our five EXC (2026-32) are deeply interdisciplinary, each providing links between at least two Excellence Areas, with EXC REC² connecting all three EAs (enlarged detail of Fig.3).

The cultivation of **EFs** will strengthen our research pipeline (cf. Fig.3). We chose four EFs at different stages of maturity: “Responsible Mobility as a Resource”, “Forest-Water Dynamics in Climate Change”, “Disruption and Resilience of Democracies”, and “Collective Adaptive Systems for Energy Supply” (cf. Annex: Tab.11). Each EF will receive tailored support to maximise their growth. We will also be preparing for the **subsequent generation of EFs** (2033-39) in areas such as space science and circular economy.

SynoSys (cf. Fig.4) has become a cornerstone of our research landscape. In the next funding phase, it will establish itself as a **global Complexity Science Hub** by introducing a visiting scholars programme, additional professorships (through reassignment), research groups, and by procuring large-scale coordinated research programmes.

RES 2: A Resourceful Science Ecosystem for Collaborative Interdisciplinary Research

We will enhance our research potential through tailored stewardship, integrating life sciences and medicine and by further strengthening DDc. Success is indicated by classical KPIs and the founding of a new School.

Our research support integrates bottom-up initiatives with strategic institutional guidance. Each School will receive **tailored research stewardship**, such as monitoring of research

outputs, identification of promising opportunities, and facilitation of interdisciplinary team formation. We will unlock the potential of our medical and life sciences by integrating them into a **new School**.

We will streamline our **TUD Dresden Fellows** programme, which invites international junior and senior scholars to TUD, to support EFs or to foster interdisciplinary collaboration. We will use **tenure-track positions** to attract ERC and DFG Emmy Noether grantees. The success of our **DDc Research Groups** in attracting and retaining talent has led us to expand this initiative to nurture EFs. We will continue to invest in **technology platforms** and include DDc partners to strengthen our research output and work on implementing **new private-public partnership models** for research infrastructure in Saxony. CIDS will serve as a blueprint for the establishment of other **Science and Resource Centres**.

EDUCATION

EDU 1: Innovative Academic Programmes Shaping Our Profile

We will enhance our educational programmes by integrating our research excellence and addressing the challenges of research-informed education in the digital age. Success is indicated by enrolments and academic success rates.

With the advent of AI and massive online learning, teaching practices must pivot to adapt. In line with Humboldt's legacy, we will evolve our academic programmes to reflect **interdisciplinarity**, and integrate **cutting-edge research** from our EXCs. A new **signature Master programme**, linked with the German Centre for Astrophysics (DZA), will focus on Astrophysics, Technology Development, and Digital Sciences. TUD will receive **5 new professorships** (Astrophysics [3], Electrical Engineering [1], Computer Science [1]) to support this programme.

We will implement a **TUD core curriculum** by educating graduates in entrepreneurship, leadership, sustainability, digital technologies and AI, and upholding democratic values to become responsible agents of change. We are strengthening research-based learning by appointing **experienced student researchers** as tutors to support professors in integrating research practices into their courses. We will expand **project-based learning** and facilitate **personalised learning pathways** by integrating AI tools into our teaching practices.

EDU 2: Educational Opportunities Throughout the Lifespan

We will develop flexible study programmes that address lifelong educational needs, while enhancing our agility in programme development. Success is reflected in student numbers and additional income.

In response to demographic ageing, we will become a university that serves **students of all ages**. This strategic shift will provide new teaching opportunities beyond the traditional early-adult student population, and address **labour-market needs** creating additional revenue streams.

To integrate scientific advances in study programmes quickly, we will create flexible study programmes by using **micro-credentials** that can be stacked toward Bachelor and Master degrees. Through close collaboration with **industry partners**, our tuition-based offerings

will address critical skills gaps in key sectors such as semiconductor technology and biotechnology.

TRANSFER

TRANS 1: Accelerating the Innovation Chain and Establishing TUD|InnoX Campuses

We will enhance our instruments to facilitate seamless technology transfer from research to application. Progress is measured using established technology-transfer KPIs.

Early industry partner integration creates resilient pathways for technology transfer that extend start-up survival. Thus, we will strengthen collaborations with leading national and international companies through strategic partnership agreements, membership in the new **excite|Club**, and the founding of **TUD|InnoX Campuses**, focusing on our EAs, that bring science (DDc partners), industry, and startups in close proximity.

To help create the region's first **unicorns**, we aim to further streamline the path from excellent research to market application. This includes: securing more EXIST fellowships and EXIST research transfer funding; expanding competitive validation grants for spin-offs; simplifying our IP policies; and improving growth conditions for spin-offs through the **Deep Tech Accelerator exciteLAB** (est. 2024), and an international scale-up programme, as part of the newly awarded **Startup Factory boOst** (BMFTR).

TRANS 2: Public Engagement for Advancing Science and Regional Welfare

We will strengthen our public engagement, science communication and leverage science and innovation to support structural transformation. Progress is measured by the number of STEM students, of citizens reached and by KPIs of structural transformation.

There is growing scepticism towards science and democratic values threatening both societal cohesion and academic freedom. Declining youth interest in STEM fields undermines future economic productivity. We will respond by further developing our **innovative dialogue formats** and leveraging **digital communication channels** to ensure our scientific insights reach and resonate with the next generation. Through **school partnerships** we will nurture democratic values and inspire interest in STEM subjects with hands-on research experiences.

TUD|Campus Lusatia leads our contributions to regional transformation, aiming to convert a former coal-mining region into an **innovation hub** focused on carbon-neutral energy supply (RefLau), circular economy (CircEcon), automated mobility (SML), and astrophysics-driven digital and technological innovation (DZA; cf. [Fig.8](#)). Through TUD|Campus Lusatia we redefine the role of academic institutions in building sustainable, resilient, and informed communities.

2. Drivers

DRIVE 1: Digitalisation Drives Institutional Transformation and Resilience

We will enhance research data analytics and cyber security to further improve our core performance areas and how we work as an institution. Progress is indicated by resilience in the face of cyber attacks, and success with data-intensive third-party funding.

We recognise the critical importance of robust digital capabilities while maintaining digital sovereignty. TUD safeguards its data and digital infrastructure while advancing international collaboration. The **maturation of CIDS** (cf. Fig.11) requires strengthening its international visibility as a digital hub for research, education, and transfer, developing the Support Centre Digitalisation (SCD) to provide unified digital services, enhanced cybersecurity measures, and streamlined administrative processes. We will expand our **capabilities in research data analytics** using advanced computational methods to derive meaningful insights from complex data sets (e.g., NFDI, EXCs).

DRIVE 2: Engineering Artificial Intelligence as a Catalyst for Institutional Excellence

We will build suitable AI tools and pursue measured implementation to enhance excellence across research, teaching, transfer, and operations. Success is reflected in research KPIs, graduation rates, technology-transfer KPIs, and cost-benefit analyses, respectively.

The emergence of AI represents an unprecedented opportunity. Our digital-technology expertise enables us to harness AI capabilities with **mathematics-based, domain-specific precision**, integrating AI to enhance our institutional capabilities beyond automation. Our goal is to integrate AI across TUD's core functions while balancing innovation with measured adoption.

DRIVE 3: Advancing Global Impact Through Transformative Partnerships

We will intensify strategic partnerships with India and Taiwan, and develop European partnerships as well as explore those with Vietnam, Singapore, and Switzerland. Impact is measured through transCampus output, academic success of international students, and start-up survival rates.

In the politically dynamic global landscape, we foster our global ecosystem of meaningful partnerships by pursuing a dual expansion strategy. We will replicate our successful **transCampus** model in at least one more location, while working with **SLOs** to enhance our research, **student recruitment**, and **technology transfer** capabilities.

Our student recruitment focuses on regions with strong academic foundations, targeting motivated youth and includes **pre-arrival exploration periods** (with TUDAG) and **employer** partnerships in Saxony. In addition to our **EUTOPIA** membership, we are strengthening our European network by intensifying partnerships with **Poland, the Czech Republic, and the Baltic States** through joint degrees, EU-funded projects, start-up activities, and the "Building Bridges Conference".

3. Enabling Resources

ENABLE 1: Attracting, Developing, and Retaining Talent

We will transform our talent management to attract, develop, and keep exceptional individuals. Success is indicated by recruitment velocity, effectiveness, and turnover rates.

In response to evolving workplace expectations and the growing regional demand for skilled professionals, we are transforming our talent management. This will include **competence mapping** to support recruitment and training interventions as well as systematic

knowledge management to buffer the ongoing generational change. We will enhance our **talent management** for TUD scientific and support staff alike to maximise their potential.

Our **competence-oriented appointment** framework accelerates hiring top professors prioritising international and female talent. **Active recruitment** (incl. tenure-track options for rising stars), an efficient and digitalised appointment process, and thoughtful onboarding take centre stage. We will **implement five professional qualification tracks** for Postdocs dedicated to science, education, research infrastructure, science management, and industry.

Central to these reforms is the creation of a modern collaborative environment transcending traditional boundaries between Divisions. By implementing **New Work practices**, we empower employees while enhancing organisational agility.

ENABLE 2: A Thriving Community as the Basis for Resilience and Agility

We will cultivate a TUD community defined by diversity, respect, integrity, trust, and open debate to meet the challenges of transformation. Success is indicated by regular staff surveys, utilisation of offerings, and number of graduates from non-academic families.

We recognise that significant change requires careful nurturing and inclusive practices. Hence, we implement programmes that develop **intercultural competencies and language skills**. Through stakeholder engagement, our school network, and mentoring programmes, we enhance access and support for **students from non-academic families**. We facilitate **staff mobility** through worldwide institutional partnerships to leverage global insights for TUD. We will foster formats of **active participation** to enrich our institutional intelligence.

By strengthening **internal communication** and creating **opportunities for interaction**, we establish the transparency vital to a vibrant community. **Sustainability practices** will be woven into our daily routines. The **200-years Jubilee** in 2028 will help us to increase visibility, strengthen community bonds, and professionalise **fundraising activities**. We will work with our **Alumni** around the world to raise our profile across Germany and beyond.

ENABLE 3: Resilient Governance for Strategic Transformation

We will advance our institutional agility through strategic foresight, evidence-informed strategic decision-making, and implementation processes that embrace collaborative engagement and transparency.

We will ensure **organisational coherence** as multiple strategies are implemented (e.g., diversity, sustainability, teaching strategy). We will strategically **allocate scarce resources** and **monitor outcomes**. We will cut red tape by leveraging the full extent of what is legally permissible to **create more room for excellence** and **blue-sky thinking**.

Our strategic governance rests on four pillars: First, we will implement **Strategic Foresight and Contextual Analysis**, recognising TUD as a complex system embedded within other systems. Second, we will facilitate required institutional transformation through implementing **change-management practices** throughout TUD (e.g., change agents). Third, we will develop **data analytics capabilities** (e.g., real-time KPI dashboard), informing strategic decision-making, planning, and monitoring at all levels. Fourth, we will improve as a **learn-**

ing organisation using systems thinking and routinely involve TUD members to contribute their perspectives.

4. Conclusion

TUD has undergone a remarkable transformation. We are a world-leading academic institution in a rapidly changing world. New technologies like the internet, AI, mobile technology, and robotics have changed how we live, learn, and generate economic productivity. At the same time, we face continuing threats to our planet's survival and new geopolitical disruptions. We aim to turn these challenges into opportunities and enter into the next phase of our institutional transformation. The science of our five EXCs is curiosity-driven but also addresses global challenges. We take responsibility for creating new opportunities and prosperity in our region, and for defending democratic values. We enhance our approach to education by moving beyond teaching knowledge and skills to preparing responsible leaders for society. We aim to reach our goals more quickly and effectively by engineering new AI tools to fit our purposes.

Pursuing our strategic goals will propel this next phase of our transformational history. We believe that deep, trusted collaboration leading to sustained engagement and inventive approaches will yield successful transformation and resilience. The stakes are high – for our planet, for democracy, and for our role as a university. Building on our proven capabilities and clear aspirations, **TUD is ready to enter a new era of Excellence.**

